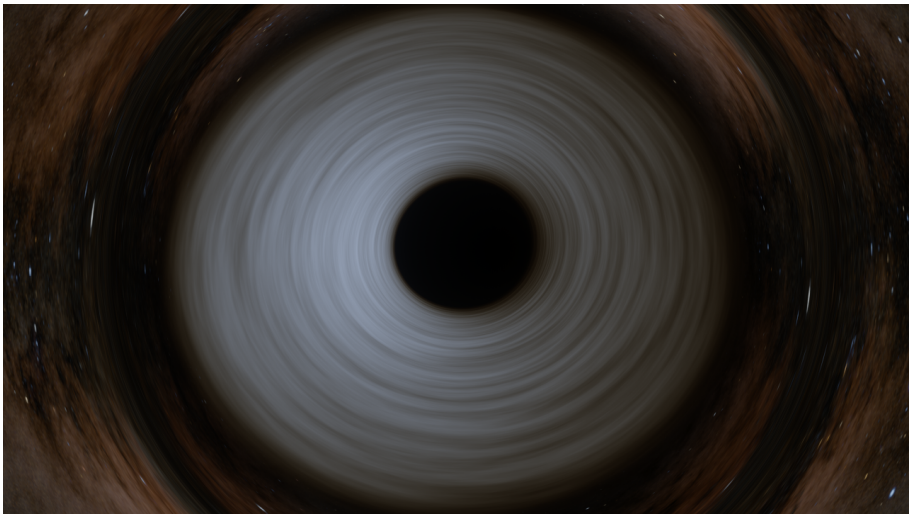
S G R A*

Extension

Seeing Sagittarius A*: true colour, moving observers and orbiting hot spots

An extension of the first-principles relativistic simulator of the Galactic-Centre black hole



C. Excelsis

Independent research report — companion to the thesis

October 4, 2026

Contents

Abstract **ix**

Relation to the thesis **xi**

Reproducibility **xii**

1 Introduction **1**

1.1 What this extension adds 1

1.2 How the new physics is validated 3

1.3 Conventions 3

1.4 Outline 4

2 Spectral Rendering: What a Camera Sees **5**

2.1 A redshifted blackbody is a blackbody 5

2.2 Visual versus bolometric Doppler contrast 6

2.3 The live colour pipeline 6

2.4 The display chain 7

2.5 A thin disk at Sgr A*'s real accretion rate 8

2.6 The lensed Milky Way 9

3 A Horizon-Penetrating Integrator **12**

3.1 The Kerr–Schild form 12

3.2 Hamiltonian equations with an analytic gradient 13

3.3 Validation 13

3.4 The live kernel 14

4 Whose Eyes? Observers in Motion **16**

4.1 Observers and their frames 16

4.2 Clocks and speeds 17

4.3 What they see . . . 18
4.4 The sky they see . . . 19
4.5 Frame dragging made visible . . . 19
4.6 Validating the live observers . . . 20

5 Orbiting Hot Spots: the GRAVITY Flares 22
5.1 The model . . . 22
5.2 Comparison with GRAVITY . . . 24
5.3 Photon-ring echoes . . . 26
5.4 The live view . . . 27

6 The Hot Accretion Flow at 230 GHz 28
6.1 The model . . . 28
6.2 The image and the EHT ring . . . 30
6.3 Polarisation . . . 32

7 Forecasts for Orbit Monitoring 36
7.1 Method . . . 36
7.2 The no-hair reach grows faster than $1/\sqrt{N}$ . . . 38
7.3 Frame dragging . . . 38
7.4 The extended mass and the no-hair parameter . . . 39

8 Falling into Sagittarius A* 42
8.1 The trip . . . 42
8.2 Time . . . 43
8.3 What you see . . . 44
8.4 Tides . . . 44
8.5 Limits . . . 45

9 Discussion and Limitations 47

10 Conclusions 49

A An Exotic Spacetime: the Traversable Wormhole 51
A.1 The metric and its rays . . . 51
A.2 The edge of the other universe . . . 52
A.3 The live demo . . . 53

B Reproducing This Report 55

List of Figures

2.1 Doppler contrast between the approaching and receding halves of the disk's major axis ($a_* = 0.94$, $i = 80°$, Carter tracer) as seen by a camera (visual luminance of $B_\nu(gT)$) and bolometrically (g^4), versus the disk's peak temperature. The shaded band is the peak temperature a thin disk would have at Sgr A*'s measured accretion rate (§2.5); dots mark its three computed values. 7

2.2 The physical preset ($a_* = 0.94$, $i = 80°$, $T_{\rm peak} = 6\,300\,\mathrm{K}$, linear display) rendered by the fp64 CPU reference (left) and the live GPU path after the colour correction (right). The approaching side, on the left, is bluer and brighter. 8

2.3 Sgr A* with a hypothetical thin disk at the computed temperature of $\dot{M} = 2 \times 10^{-8}\,\mathrm{M_\odot\,yr^{-1}}$. Top: the EHT viewing geometry ($i = 30°$, left) and the physical preset ($i = 80°$). Bottom left: a close orbit with the cloud layer. Bottom right: the physical preset in the bolometric view, which shows the g^4 beaming in grey. The background is the lensed Milky Way (§2.6). Engine commands: `scripts/render_ext_gallery.py`. 10

4.1 One place, three observers: the close-orbit scene seen by a ZAMO (left), by an observer falling from rest at infinity (centre) and by one orbiting at the Keplerian rate (right). Only the camera's tetrad differs. 18

4.2 The shadow along the image's horizontal axis for each observer and camera position (bars: fp64 Kerr–Schild bisection; ticks: the live GPU renderer). At $a_* = 0$ the bars equal Synge's shadow aberrated by the observer's velocity. 19

4.3 Light that passed through the ergoregion and escaped (blue overlay) around Sgr A* seen nearly edge-on ($i = 88°$, $r = 30\,M$, no disk; lensed Milky Way behind). Captured light is not marked. The crescent appears only on the prograde side, where photon orbits dip inside the ergosphere. 20

5.1 Hot spots on the GRAVITY orbits around Sgr A*. Left: light curves over one orbit, normalised to the mean (dotted: the part carried by lensed images). Right: the flux-weighted centroid tracks on the sky (dot: start; arrows: sense of motion; cross: the black hole). The grey bar shows the centroid motion GRAVITY measured in the July 2018 flare, about $120\,\mu$as in about 30 minutes (GRAVITY Collaboration et al., 2018). 25

5.2 Photon-ring echoes on the GRAVITY 2018 orbit. Left: a brief flash ($\tau = 1\,M$) seen in the direct image and, delayed, in the first lensed image (scaled up tenfold). Right: the echo delay versus the orbital phase of the flash for the three orbits of Table 5.1. 26

5.3 The GRAVITY 2018 hot spot around Sgr A* ($R = 7\,M$, $i = 160°$). Top: fp64 images (logarithmic stretch, four decades) at quarter orbits and averaged over one orbit; the thin arcs near the shadow edge are the lensed images. Bottom: the live renderer at the same phases from $45\,M$, with the lensed Milky Way behind and the spot in false colour. 27

6.1 Sgr A* at 230 GHz in the RIAF model ($a_* = 0.94$, $i = 30°$, n_0 fitted to 2.4 Jy), as brightness temperature. Left: the model image. Right: blurred with the EHT's $20\,\mu$as restoring beam. Dashed white: the EHT ring diameter, $51.8\,\mu$as (Event Horizon Telescope Collaboration, 2022); cyan: the ring measured on the model image. 30

6.2 Visibility amplitudes of the model image of Fig. 6.1 along twelve baseline position angles (blue) and averaged over all (black). Dotted: the minima of the EHT data (Event Horizon Telescope Collaboration, 2022). A beam does not move these minima. 31

6.3 One camera ($i = 30°$, $60\,M$), three bands. Left: visible light, with the hypothetical thin disk of Chapter 2 and the lensed Milky Way. Centre: 230 GHz, the hot accretion flow of this chapter (brightness temperature, false colour). Right: 2.2 μm, a GRAVITY hot spot (Chapter 5; false colour). Only the 230 GHz view shows Sgr A*'s real quiescent flow. 32

6.4 EVPA patterns of the 230 GHz RIAF for three magnetic-field geometries, with the flow rotating clockwise on the sky ($i = 150°$), blurred with the EHT beam. Grey: total intensity. Ticks: EVPA, length proportional to the polarised intensity, colour to the fractional polarisation. Only the vertical (poloidal) field gives the spiral seen by the EHT. 35

7.1 Left: forecast uncertainty of the no-hair parameter versus monitoring time after 2017.0, for the two continuations of the data (grey lines: S2's pericentre passages). The dashed curve is the thesis' $1/\sqrt{N}$ extrapolation; the dots compare it with the Fisher forecast at ten more orbits. Right: significance of the spin of Sgr A* ($a_* = 0.94$, Lense–Thirring and Kerr quadrupole) for three astrometric precisions per coordinate (green; purple: with an extended mass fitted at the same time). Spin axis along the line of sight. . . 39

7.2 Left: joint 1σ constraints on the extended mass enclosed within S2's apocentre and the no-hair parameter, after 8, 16 and 32 yr of GRAVITY-era monitoring (grey: GRAVITY's present 1σ bound). Right: the false no-hair signal, in units of the forecast $\sigma(\varepsilon_q)$, when the extended mass is left out of the fit. 41

8.1 Falling into Sgr A* ($a_* = 0.94$): the view straight ahead of an observer dropped from rest at infinity, at polar angle 60°, live GPU renders of the double-precision worldline (field of view 1.6 rad; the lensed Milky Way is the only light). Top: outside the horizon ($r_+ = 1.341\,M$); bottom: just outside it and inside it. Labels give the radius and the faller's proper time since $r_0 = 30\,M$. 45

8.2 The faller's instruments along the trip (dashed: the horizon; dotted: the ergosphere at this latitude). Left: its proper time and the far-away time, which diverges at r_+. Middle: frequency of the sky light arriving from straight behind and from the side, relative to its frequency at infinity. Right: tidal stretch across a 2 m body. . . . 46

A.1 *Not Sgr A*:* the Interstellar wormhole rendered live (GPU), approached along ℓ (field of view 1.2 rad). The fp64 CPU tracer's results are overlaid: the edge of the other universe (dashed; eq. A.4) and the Einstein ring (dotted). Inside the edge is the other universe (an illustrative recoloured sky); outside it is the lensed Milky Way, with the Galactic Centre behind the throat. 54

List of Tables

1.1 Phenomena added by the extension. "Thesis" marks what the thesis already modelled; chapters in brackets are pending. 2

2.1 The live renderer's scenes: one spacetime ($a_* = 0.94$), five cameras and accretion rates. The thin disk is hypothetical (§2.5); its peak temperature follows from $\dot{M}$. "Soft" styles fade the disk's outer rim; "clouds" adds an optically thin, locally thermal cloud layer moving on the disk's own circular orbits (display only, removable). 9

4.1 Speed v relative to the local ZAMO [c] and clock rate $d\tau/dt$ (observer proper time per unit far-away time) of the four standard observers on the equator (`observer_tool`). 18

5.1 Simulated observables of an orbiting hot spot around Sgr A* ($a_* = 0.94$; $s = 0.5\,M$), from `flare_tool`. Loop: the largest separation of two centroid positions; in brackets the geometric orbit diameter. Chord: the largest centroid displacement within 30 minutes and the corresponding speed. Contrast: maximum over minimum flux for $\alpha = 0$ ($\alpha = 0.6$). Lensed: mean flux fraction in images of order $n \geq 1$. Echo: delay of the $n = 1$ image of a flash behind the direct one, over eight orbital phases. 24

6.1 The RIAF at 230 GHz: fitted density normalisation, ring diameter on the model image, first visibility minimum (shortest over position angles; in brackets the equivalent thin-ring diameter) and brightness asymmetry after the 20 μas beam, for the reference model and variations (each refitted to 2.4 Jy). EHT values from Event Horizon Telescope Collaboration (2022). 32

6.2 Polarisation of the RIAF at 230 GHz for three field geometries, seen with counter-clockwise ($i = 30°$) and clockwise ($i = 150°$) rotation on the sky: phase of β_2 on the model and blurred images (degrees), and on the blurred image $|\beta_2|$, $|\beta_2|/|\beta_1|$, the resolved $\langle|m|\rangle$ and the net $|m_{\rm net}|$. $|\beta_2|$, $\langle|m|\rangle$ and $|m_{\rm net}|$ scale with the assumed intrinsic fraction $p_0 = 0.7$. EHT 90 per cent ranges from Event Horizon Telescope Collaboration et al. (2024a). 34

7.1 An extended mass (Plummer, scale $0.3''$) present in the data but left out of the fit: biases of the 13-parameter Kerr fit in M and R_0 and the χ^2 it leaves, and the false no-hair signal in the thesis' ε_q profile. Noise-free data on the Gillessen et al. (2017) epochs, plus the stated years of GRAVITY-era monitoring. For M and R_0, σ is the forecast uncertainty of the same data set; for ε_q it is the uncertainty the biased fit itself would quote, from the curvature of its profile. 41

A.1 The Dneg wormhole of *Interstellar* ($2a = 0.01\,\rho$, $W = 0.05\,\rho$) seen from three distances: angular radius of the other universe (CPU, equal to eq. A.4) and of the Einstein ring. The live GPU edges agree with the CPU edges within 0.28 pixels in all three cases. . . 53

Abstract

This report extends the Sgr A* simulator of the accompanying thesis from measuring the black hole to *seeing* it: every image is now rendered as a camera would record it, from any observer's eyes, with time-dependent sources — and every new number is again produced by tested code. All scenes use one spacetime, Sgr A* with $a_* = 0.94$ and the GRAVITY mass and distance.

(i) *Spectral rendering.* A redshifted blackbody is a blackbody at gT, so one Planck–CIE lookup carries Doppler colour and beaming. The live GPU renderer now reproduces the fp64 CPU reference pixel for pixel (median luminance difference 0.000, 95th percentile 2.0 %; Doppler contrast within 0.26 %). A thin disk fed at Sgr A*'s measured accretion rate would peak at 7 280–23 000 K and shine 12–1 200 times brighter than Sgr A* actually does — a quantitative reason its real accretion flow cannot be a thin disk. A camera's Doppler contrast differs from the bolometric g^4 law: stronger below $\approx 11\,500$ K, weaker above. (ii) *A horizon-penetrating integrator.* Cartesian Kerr–Schild Hamiltonian geodesics with an analytic gradient replace the Boyer–Lindquist live kernel, removing its polar seam. (iii) *Observers.* Static, orbiting, free-falling and hovering observers see the aberrated, Doppler-shifted sky predicted by their Lorentz- boosted tetrads; the Schwarzschild shadow edges match Synge's formula plus special-relativistic aberration to 2×10^{-12}, and the live renderer matches the fp64 reference to 0.72 pixel. The comparison exposed, and fixed, a step-budget defect that had widened the prograde edge of the high-spin shadow by 16.8 pixels. (iv) *Flares.* A Gaussian hot spot on the orbit GRAVITY fitted to the 2018 near-infrared flares ($R = 7\,M$, $i \approx 160°$), traced with slow light, has a period of 43.1 min (observed 40 ± 8 min) and moves clockwise as observed, but its centroid travels only 64 μas in 30 minutes against the observed $\sim$120 μas — reproducing, independently, the tension GRAVITY reported between flare periods and loop sizes. Each flash returns as a photon-ring echo 5.9–9.1 minutes later. (v) *The hot flow at 230 GHz.* A semi-analytic radiatively inefficient flow, normalised to the EHT's 2.4 Jy and

traced with thermal-synchrotron radiative transfer, puts its light in a Doppler-boosted crescent inside the photon ring. Its ring, $42.8\,\mu$as, and its first visibility minimum, $3.90\,\mathrm{G}\lambda$, are smaller than the EHT's $51.8\,\mu$as and $3.0\,\mathrm{G}\lambda$: the emission must sit further out, for instance in cooler electrons. Transported to the camera with the Walker–Penrose constant, its polarisation shows a radial pattern for a toroidal field, which the EHT excludes. Only a vertical, poloidal field produces the observed spiral ($\angle\beta_2 = -135°$ for clockwise rotation).

(vi) *Forecasts for S2.* Fisher forecasts from the thesis' own forward model, checked against its profile likelihood, show that the no-hair constraint tightens much faster than the thesis' $1/\sqrt{N}$ extrapolation. Ten more orbits of existing quality would give $\sigma(\varepsilon_q) = 577$, not 34 000, because ε_q acts through a secular precession. An extended mass at GRAVITY's present bound, left out of the fit, becomes a 5.0σ false no-hair signal after 32 years of GRAVITY-quality data, invisible in the fit's χ^2. Fitting it jointly costs only a factor 1.23. Frame dragging in S2's orbit needs 34 years of $10\,\mu$as astrometry for a 3σ detection. (vii) *Falling into Sgr A*.* A timelike geodesic in the ingoing Kerr–Schild chart carries the camera through the event horizon, with its conserved quantities kept to 3×10^{-13}. From $30\,M$, the faller reaches the horizon after 27.0 minutes of proper time, and the inner horizon, where the model stops, 8.5 s later. The tidal stretch across a 2 m body is $3.7 \times 10^{-3}\,\mathrm{m\,s^{-2}}$ at the horizon. Ahead, the sky skimming the dark region is blueshifted up to 27.3-fold. The live view matches the double-precision tracer to 0.73 pixel, inside the horizon as well as outside.

An appendix applies the renderer to an exotic spacetime that is *not* Sgr A*, the traversable wormhole of *Interstellar*, as a test of its generality.

Relation to the thesis

This is a companion report, deliberately kept separate from the thesis. It reuses the thesis' physics library, validation conventions and calibration (e.g. one gravitational radius subtends 5.1243 μas and one unit of geometric time is 0.3527 minutes at the GRAVITY mass and distance), and it cites the thesis' chapters rather than repeating them. Two thesis results are prerequisites:

- the image-handedness correction (thesis Appendix "Image handedness: an error found and corrected"): every camera here traces the photon it *receives*; images are not mirrored;
- the validated Carter/Mino science path, which remains the reference against which every new renderer in this report is tested.

Display-only processing (exposure, bloom, white balance, tone curve, false colour) is labelled wherever it appears and never enters a measurement.

Reproducibility

Every number in this report is a macro in `numbers_ext.tex`. The macros are written by the script `make_numbers_ext.py` (in `scripts/`) from the JSON and CSV outputs of the tools listed in Appendix B. Every figure is generated by `make_figures_ext.py` or rendered by a documented engine command. `make reproduce-ext` regenerates all of them.

Chapter 1

Introduction

The thesis built a simulator of Sgr A* to *measure* things: the size and shape of the shadow and photon ring against the Event Horizon Telescope (Event Horizon Telescope Collaboration, 2022), the mass and distance from the orbit of S2 (Gillessen et al., 2017; GRAVITY Collaboration, 2018, 2020, 2022), and a free-quadrupole test of the no-hair theorem. Its images were instruments — bisected boundaries, sub-ring decompositions, flux ratios — and the renders that accompanied them were illustrations.

This report asks a different question of the same spacetime: *what would one actually see?* Answering it honestly takes physics the thesis did not need. A camera does not record bolometric intensity but a spectrum filtered by its colour response. A moving observer sees a different sky from a static one at the same point. The light reaching the camera now left its source at different times along different paths, so a variable source — the near-infrared flares of Sgr A* — appears smeared, delayed and echoed. And the real accretion flow of Sgr A* is not the thin disk of the thesis' renders at all, but a hot, optically thin plasma best seen at millimetre wavelengths.

1.1 What this extension adds

Table 1.1 lists the phenomena the simulator now shows, the gate of the extension plan that added each one and the chapter that documents it. Every scene uses one spacetime — Sgr A* with $a_* = 0.94$ (Event Horizon Telescope Collaboration, 2022) and the GRAVITY mass and distance (GRAVITY Collaboration, 2022). The only exception, a traversable wormhole, appears in an appendix as a labelled test of the code's generality, never as a model of Sgr A*.

Table 1.1: Phenomena added by the extension. "Thesis" marks what the thesis already modelled; chapters in brackets are pending.

Phenomenon	Gate	Chapter
Shadow, lensing, photon sub-rings, g^4 beaming, S2 orbit in full GR, no-hair test	thesis	—
True visual colour of a redshifted blackbody; Doppler hue shift	G13	2
High-dynamic-range display; a thin disk at Sgr A*'s measured accretion rate	G14	2
Lensed real Milky Way panorama	G15	2
Horizon-penetrating Kerr–Schild integrator	G16	3
Moving observers: aberration, sky Doppler shift, time dilation, frame dragging, light that visited the ergoregion	G17	4
Orbiting hot spots (the GRAVITY flares) with slow light and photon-ring echoes	G18	5
Hot accretion flow at 230 GHz; multi-band views	G19	6
Polarisation	G20	6
Forecasts: extended-mass bias, Lense–Thirring detectability	G22	7
A fall through the horizon of Sgr A*	G24	8
Traversable wormhole (exotic, not Sgr A*)	G21	App. A

1.2 How the new physics is validated

The extension keeps the thesis' rule that a picture is not evidence. Each new capability follows the same pattern:

1. a *CPU twin* in double precision, built on the thesis' physics library, produces the reference numbers;
2. *tests* pin the twin to closed forms wherever one exists — the Schwarzschild limit, Bardeen–Press–Teukolsky circular orbits (Bardeen et al., 1972), Synge's shadow (Synge, 1966), the Carter constants (Carter, 1968), exact symmetries;
3. the *live GPU path* (single precision, real time) is compared with the twin pixel by pixel, and the agreement is quoted;
4. every number in this report is written by a tool and reaches the text through a generated macro.

The comparisons in step 3 are not formalities. The observer comparison of Chapter 4 exposed a defect in the live renderer that had shifted the prograde edge of the high-spin shadow by 16.8 pixels. While building the slow-light test of Chapter 5, a reimplementation of the Carter integrator without the thesis' constraint projection disagreed with the Kerr–Schild path at the 10^{-4} level, which showed that the projection matters even at moderate inclinations.

1.3 Conventions

Units are geometric, $G = c = M = 1$, unless stated. At the GRAVITY mass and distance one gravitational radius subtends $5.1243\,\mu$as and one unit of coordinate time lasts 0.3527 minutes. Cameras trace the photon they *receive* (thesis Appendix on image handedness); the default camera looks at the hole with image-right along the local $+\hat{\phi}$ direction and image-up toward the spin axis, so prograde material approaches on the left. Inclinations are measured from the spin axis; $i > 90^\circ$ views the hole from below its equatorial plane, where prograde motion appears clockwise.

1.4 Outline

Chapter 2 treats colour and brightness: what a camera sees of a redshifted blackbody, the corrected live colour pipeline, the display chain, the hypothetical thin disk fed at Sgr A*'s real accretion rate, and the lensed Milky Way. Chapter 3 describes the Kerr–Schild integrator that now drives every live image. Chapter 4 adds observers in motion. Chapter 5 models the GRAVITY flares as orbiting hot spots and compares them with the observations. Chapter 6 replaces the hypothetical thin disk by Sgr A*'s real, hot accretion flow at 230 GHz and compares its total-intensity and polarised images with the EHT's. The remaining chapters — forecasts for future orbit monitoring and a fall into Sgr A* — will follow their gates.

Chapter 2

Spectral Rendering: What a Camera Sees

The thesis measured beaming with the bolometric intensity, which scales as g^4 and is the right quantity for flux ratios. A camera records something else: the spectrum weighted by its colour response. This chapter derives what that changes, documents the live renderer's colour pipeline (and the defect found in it), and uses the result to ask what Sgr A* would look like if it were fed by a thin disk at its measured accretion rate.

2.1 A redshifted blackbody is a blackbody

Along a light ray in vacuum I_ν/ν^3 is conserved (e.g. Rybicki & Lightman, 1979; Misner et al., 1973). Light emitted at frequency ν_e and received at $\nu = g\nu_\mathrm{e}$ therefore arrives with $I_\nu = g^3 I^{(\mathrm{e})}_{\nu/g}$. For a blackbody emitter, $I^{(\mathrm{e})}_\nu = B_\nu(T) = (2h\nu^3/c^2)/(e^{h\nu/kT}-1)$, and

$$g^3\, B_{\nu/g}(T) = g^3\, \frac{2h(\nu/g)^3/c^2}{e^{h\nu/(gkT)}-1} = B_\nu(gT). \tag{2.1}$$

The received spectrum is exactly a blackbody at the observed temperature $T_\mathrm{obs} = gT$. One lookup table of $B_\nu(T)$ through the CIE 1931 colour-matching functions (fitted as in Wyman et al., 2013) then carries everything at once: the Doppler and gravitational colour shift and the beaming. Only the *bolometric* integral of (2.1) is the g^4 law, $\int B_\nu(gT)\,d\nu = \sigma(gT)^4/\pi$. A camera integrates $B_\nu(gT)$ against its own spectral response instead, and that does not scale as any single power of g.

2.2 Visual versus bolometric Doppler contrast

The difference matters most where it is largest, between the approaching and receding sides of the disk. Along the image major axis of the thesis' reference view ($r_{\mathrm{cam}} = 1000\,M$, $i = 80°$, $a_* = 0.94$, disk to $18\,M$; the Carter science tracer) the redshift factor of the disk ranges from $g = 0.168$ on the receding limb to $g = 1.309$ on the approaching one. `spectral_tool` sums both the bolometric intensity and the visual luminance Y over the two halves of that axis (Fig. 2.1).

The bolometric ratio, 52.2, does not depend on the disk temperature: every point scales by the same T^4. The visual ratio depends on it strongly. For the thesis' reference temperature of 6300 K a camera sees the approaching side 1 090 times brighter than the receding one, 21 times the bolometric contrast. The receding limb's light is shifted out of the visible band onto the exponential Wien tail. For hot disks the visible band sits on the Rayleigh–Jeans side, where $B_\nu \propto T$, and the visual contrast drops below the bolometric one. The two are equal at $T \approx 11\,500$ K. The colour shifts with them: at 6300 K the hottest annulus seen at the extreme Doppler factors has chromaticity $(x, y) = (0.293, 0.302)$ on the approaching side, a blue-white, and $(0.635, 0.361)$ on the receding side, deep orange.

2.3 The live colour pipeline

The live renderer (Phases 8–15 of the thesis) coloured the disk with an 8-bit lookup table computed from the same Planck integration. Reading the code for this extension revealed three problems:

1. the table stored *radiance-weighted* colours,
2. it was gamma-encoded and uploaded as normalised integers,
3. the shader then multiplied by $(T_{\mathrm{obs}}/6500\,\mathrm{K})^4$ again.

The temperature dependence was therefore counted twice. A 24-stop luminance grade then mapped the result onto a narrow beige palette, which removed the Doppler colour shift that (2.1) predicts. A separate error in the shader's Page–Thorne flux, an extra factor $1/x^2$, made the live disk fall off as r^{-4} instead of r^{-3} and run cooler than the temperature it was given. The science path, which colours its figures with the CPU function `planck_to_linear_srgb`, was not affected.

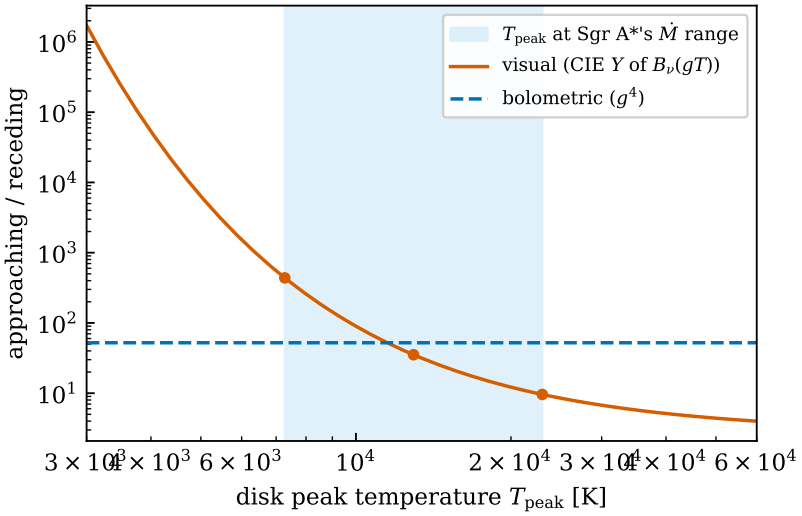


Figure 2.1: Doppler contrast between the approaching and receding halves of the disk's major axis ($a_* = 0.94$, $i = 80°$, Carter tracer) as seen by a camera (visual luminance of $B_\nu(gT)$) and bolometrically (g^4), versus the disk's peak temperature. The shaded band is the peak temperature a thin disk would have at Sgr A*'s measured accretion rate (§2.5); dots mark its three computed values.

The corrected pipeline builds a floating-point table of $B_\nu(T)$ through the CIE curves at start-up from that same CPU function: 1024 entries, logarithmic in T from 300 K to 10^5 K, normalised so that a 6500 K blackbody has luminance 1. The shader makes one lookup at gT and nothing else. A bolometric display mode, which shows $(gT/6500\,\mathrm{K})^4$ in grey, is kept for teaching the g^4 law.

The validation (`compare_live_colour.py`) renders the thesis' physical preset twice at identical parameters ($T_{\mathrm{peak}} = 6\,300$ K): once on the GPU and once with the CPU tracer and the CPU shading. Over 24 728 comparable disk pixels the median luminance difference is $|\Delta \ln Y| = 0.000$ (95th percentile 2.0 %), and the median chromaticity difference is 0.000. The approaching-to-receding luminance ratio is 17.9 on the CPU and 18.0 on the GPU, a relative difference of 0.26 %. The two disk footprints differ in 0.33 % of the image's pixels, all at the one-pixel boundary between single- and double-precision tracing (Fig. 2.2).

2.4 The display chain

What a screen can show is not what the radiance image contains. Between the two sits a display chain, applied *after* the physics and switched off completely in

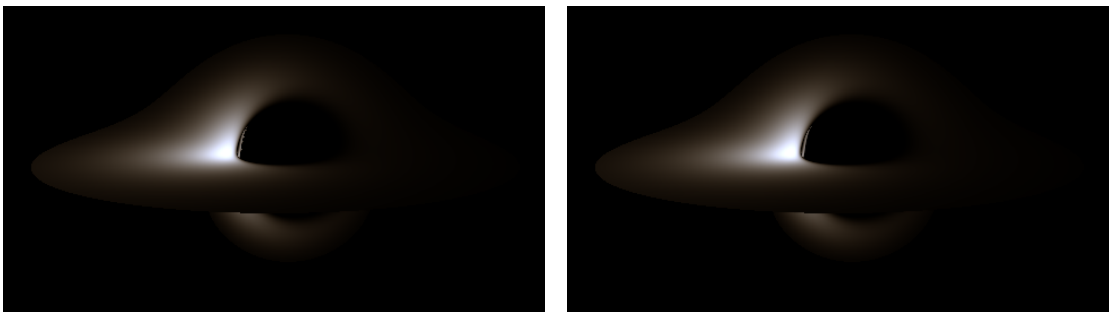


Figure 2.2: The physical preset ($a_* = 0.94$, $i = 80°$, $T_{\mathrm{peak}} = 6\,300\,\mathrm{K}$, linear display) rendered by the fp64 CPU reference (left) and the live GPU path after the colour correction (right). The approaching side, on the left, is bluer and brighter.

Science Mode, which writes the linear image with the same transform as the CPU reference:

Progressive accumulation. While the view is static, every frame adds one jittered sample per pixel to a floating-point running mean. The jitter follows the R2 low-discrepancy sequence (Roberts, 2018). Sample 0 is the un-jittered pixel centre, so one sample reproduces the historical image exactly.

Exposure metering. Exposure value 0 places the luminance of the un-beamed disk at its peak temperature at display level 1, as a camera's meter would, so disks of 7000 K and 23 000 K are both well exposed.

Bloom. An energy-conserving dual-filter mip pyramid (Jimenez, 2014), with a luminance-weighted first level so that single hot pixels cannot flicker into glare.

White balance and tone curve. White-balance gains render a blackbody of the chosen temperature neutral. A filmic curve, AgX (Sobotka, 2022, via the polynomial fit of Wrensch 2023) or ACES (Narkowicz, 2016), compresses the range while keeping hue in overexposed regions.

2.5 A thin disk at Sgr A*'s real accretion rate

The thesis' disk temperature was a display choice. Here it is computed. Restoring G and c in the Page–Thorne flux (Page & Thorne, 1974) gives $F(r) = 3\dot{M}c^6/(8\pi G^2 M^2)\, f(r)$ and $\sigma T_{\mathrm{eff}}^4 = F$. The accretion rate of Sgr A* is bounded by

Table 2.1: The live renderer's scenes: one spacetime ($a_* = 0.94$), five cameras and accretion rates. The thin disk is hypothetical (§2.5); its peak temperature follows from $\dot{M}$. "Soft" styles fade the disk's outer rim; "clouds" adds an optically thin, locally thermal cloud layer moving on the disk's own circular orbits (display only, removable).

Scene	style	i [deg]	$r_{\rm cam}/M$	$r_{\rm out}/M$	$\dot{M}$ [$\mathrm{M_\odot\,yr^{-1}}$]	$T_{\rm peak}$ [K]
Physical	exact	80	60	18	2×10^{-8}	12 900
Earth View ($i = 30°$)	soft	30	60	18	2×10^{-8}	12 900
Close Orbit	soft, clouds	84	45	24	2×10^{-8}	12 900
Low $\dot{M}$	soft	78	60	18	2×10^{-9}	7 280
High $\dot{M}$	soft	78	60	18	2×10^{-7}	23 000

the Faraday rotation of its 230 GHz emission to $2\times10^{-9} \lesssim \dot{M} \lesssim 2\times10^{-7}\,\mathrm{M_\odot\,yr^{-1}}$ (Marrone et al., 2007); see also the review by Genzel et al. (2010). Over that range a Novikov–Thorne disk around Sgr A* ($a_* = 0.94$) would peak at

$$T_{\rm peak} = 7\,280\text{–}23\,000\ \mathrm{K} \qquad (\dot{M} = 2\times10^{-9}\text{–}2\times10^{-7}\ \mathrm{M_\odot\,yr^{-1}}), \tag{2.2}$$

at $r = 2.95\,M$ (12 900 K at 2×10^{-8}). It would look blue-white: the chromaticity at its peak moves from $(0.303, 0.313)$ to $(0.254, 0.254)$ across the range.

Such a disk would also be far too bright. Its luminance $L = \eta\dot{M}c^2$, with radiative efficiency $\eta = 1 - E_{\rm ISCO} = 0.181$ for this spin, spans 2.1×10^{37}–$2.1\times10^{39}\,\mathrm{erg\,s^{-1}}$, or 3.8×10^{-8}–3.8×10^{-6} of the Eddington luminosity ($5.4\times10^{44}\,\mathrm{erg\,s^{-1}}$ at the GRAVITY mass). Sgr A* radiates on average about $10^{-8.5}$ of its Eddington luminosity (Genzel et al., 2010). A thin disk fed at the measured rate would therefore outshine it by a factor 12–1 200. This is the classic argument, made quantitative here, that the accretion flow of Sgr A* is radiatively inefficient: a hot, geometrically thick, optically thin plasma that radiates a tiny fraction of its accretion energy. That flow is the subject of Chapter 6. The thin disk survives in the renderer as a labelled hypothetical, with its temperature computed rather than chosen (Table 2.1, Fig. 2.3).

2.6 The lensed Milky Way

The thesis rendered a procedural starfield. The background is now the real sky: the NASA SVS *Deep Star Maps 2020* panorama (Wright, 2020), built from Gaia

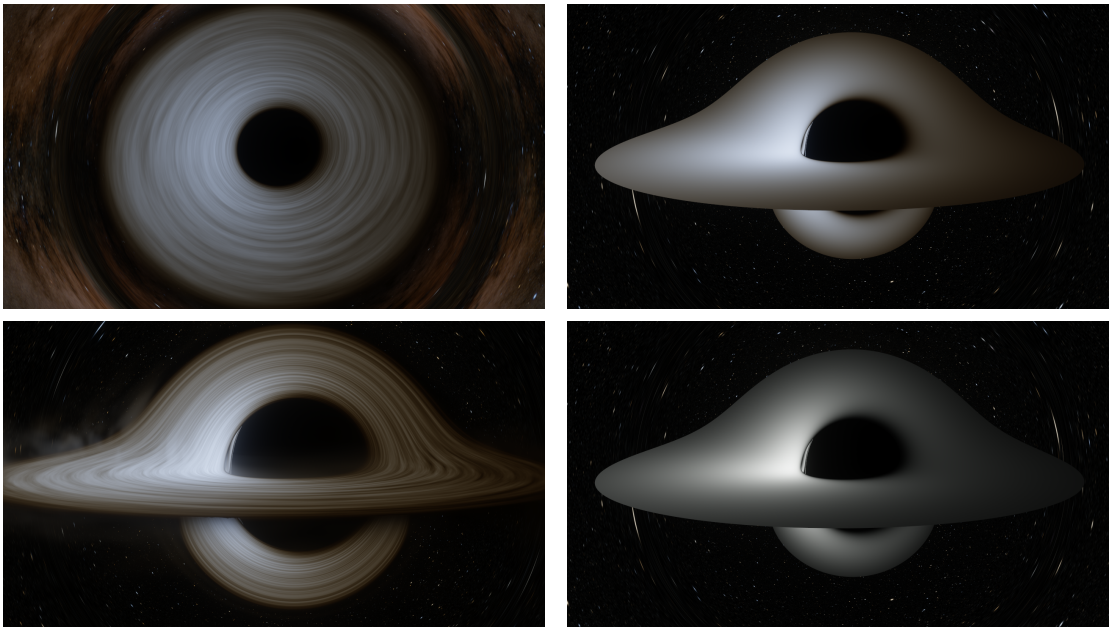


Figure 2.3: Sgr A* with a hypothetical thin disk at the computed temperature of $\dot{M} = 2 \times 10^{-8}\,\mathrm{M}_\odot\,\mathrm{yr}^{-1}$. Top: the EHT viewing geometry ($i = 30°$, left) and the physical preset ($i = 80°$). Bottom left: a close orbit with the cloud layer. Bottom right: the physical preset in the bolometric view, which shows the g^4 beaming in grey. The background is the lensed Milky Way (§2.6). Engine commands: `scripts/render_ext_gallery.py`.

DR2 (Gaia Collaboration et al., 2018), Hipparcos-2 and Tycho-2, in galactic coordinates (plate carrée, 8192×4096). Each escaping ray reads the map in its asymptotic direction. The texture level follows the pixel footprint, so the dense star clouds do not alias; strongly lensed regions alias per sample, which progressive accumulation averages away. The map is placed with the spin axis at $i = 30°$ to our line of sight (Event Horizon Telescope Collaboration, 2022), so a camera at that inclination sees the Galactic-Centre star clouds behind the hole. The spin's position angle on the sky is not constrained by current data and is set to zero, an assumption stated in the interface.

Two caveats apply. The panorama is the sky seen *from Earth*, used as a backdrop at infinity: from the Galactic Centre, stars within a few kiloparsecs would lie in different directions. And the map is an RGB image, not a spectrum: when an observer's motion or gravity shifts the received light by a factor g (Chapter 4), each map pixel is shifted as a 6500 K blackbody would be, $B(6500\,\mathrm{K}\cdot g)/B(6500\,\mathrm{K})$ per channel, which gives the right sense and size of the colour and brightness change but not a star-by-star spectrum.

Chapter 3

A Horizon-Penetrating Integrator

The thesis integrates Carter's separated equations in Boyer–Lindquist (BL) coordinates, in Mino time, in double precision. For metrology that is ideal: the constants of motion are exact, and the integrator is tested to 10^{-12}. For a real-time renderer the same coordinates bring two problems. The polar axis $\theta = 0, \pi$ is a coordinate singularity. Rays passing near it needed clamps, a snap of the angular momentum and a step-squeeze heuristic, and they still left a faint seam down the image's central column. And the horizon is a coordinate singularity too ($\Delta \to 0$), where backward-traced captured rays stall. This chapter describes the replacement used by every live image of the extension: Hamiltonian geodesics in Cartesian Kerr–Schild coordinates (Kerr & Schild, 2009; Visser, 2007).

3.1 The Kerr–Schild form

In Kerr–Schild form the Kerr metric is a flat metric plus a null-vector term (Kerr, 1963; Kerr & Schild, 2009),

$$g_{\mu\nu} = \eta_{\mu\nu} + f\, l_\mu l_\nu, \qquad f = \frac{2Mr^3}{r^4 + a^2 z^2}, \tag{3.1}$$

$$l_\mu = \left(1,\ s\,\frac{rx + s\,ay}{r^2 + a^2},\ s\,\frac{ry - s\,ax}{r^2 + a^2},\ s\,\frac{z}{r}\right), \tag{3.2}$$

where the BL radius r is the positive root of $r^4 - (x^2 + y^2 + z^2 - a^2)\, r^2 - a^2 z^2 = 0$ and $s = \pm 1$ selects the chart. Because l is null for both η and g, the inverse needs no matrix inversion: $g^{\mu\nu} = \eta^{\mu\nu} - f\, l^\mu l^\nu$. The relation to BL coordinates is

$$\begin{aligned} x + iy &= (r + isa)\sin\theta\, e^{i\phi_{\mathrm{KS}}}, \qquad z = r\cos\theta, \\ t_{\mathrm{KS}} &= t_{\mathrm{BL}} + s\,T(r), \qquad \phi_{\mathrm{KS}} = \phi_{\mathrm{BL}} + s\,\Phi(r), \end{aligned} \tag{3.3}$$

with $T = \int 2Mr/\Delta\, dr$ and $\Phi = \int a/\Delta\, dr$. The two charts differ in which horizon they cover regularly. The *outgoing* chart ($s = -1$) is regular across the past horizon, which is where camera rays traced backward in time end when they fall in. The *ingoing* chart ($s = +1$) is regular across the future horizon, which an in-falling observer crosses (Chapter 8).

3.2 Hamiltonian equations with an analytic gradient

With $H = \frac{1}{2} g^{\mu\nu} p_\mu p_\nu$ and $l \cdot p \equiv l^\mu p_\mu$, the flat part of the inverse metric is constant, so Hamilton's equations reduce to

$$\frac{dx^\mu}{d\lambda} = \eta^{\mu\nu} p_\nu - f\, l^\mu\, (l \cdot p), \qquad \frac{dp_i}{d\lambda} = \tfrac{1}{2}\, \partial_i \big[f\, (l \cdot p)^2 \big], \qquad \frac{dp_t}{d\lambda} = 0. \tag{3.4}$$

The gradient follows in closed form from $\partial_i r = r(r^2 x_i + a^2 z\, \delta_{iz})/(r^4 + a^2 z^2)$ and the explicit f and l. Unlike the finite-difference gradients common in real-time renderers, this introduces no step-size-dependent truncation error. Stationarity and axisymmetry make $E = -p_t$ and $L_z = x\, p_y - y\, p_x$ exact constants. The Carter constant is recovered by transforming the covector back to BL components, and $H = 0$ measures the null condition.

The double-precision twin (`physics/src/kerr_schild.cpp`) integrates (3.4) with the thesis' adaptive Dormand–Prince scheme. Cameras build their rays in BL components in the observer's frame (Chapter 4), map them to Kerr–Schild components with the Jacobian of the coordinate change, and lower them with (3.1).

3.3 Validation

The Catch2 suite (`test_kerr_schild.cpp`) checks the following:

- (3.1) reproduces the BL Kerr metric, transformed, to 10^{-12} at 240 points;
- the inverse and the analytic gradient match their numerical counterparts;
- the Schwarzschild limit gives the critical impact parameter $\sqrt{27}\, M$;
- the constants of motion are conserved;

- the camera traces the photon it receives (thesis Appendix on image handedness).

Against the thesis' Carter tracer the image-plane comparison of the thesis appendix gives, over 24 pixels off the spin-axis column:

- disk-hit radii agree to 2.7×10^{-11} and redshift factors to 1.8×10^{-11};
- the worst drifts along the Kerr–Schild rays are 6.1×10^{-12} in L_z, 7.7×10^{-12} in Q and 6.6×10^{-11} in H, far inside the thesis' 10^{-6} health criterion.

On the image's central column the agreement is 6.4×10^{-5}. That column is where BL rays pass closest to the polar coordinate singularity, so the difference measures the reference, not the new integrator.

The time coordinate needs its own check once sources vary (Chapter 5). Along a Kerr–Schild ray $dt/d\lambda = -p_t + f\,(l \cdot p)$. In the outgoing chart the KS time of an equatorial crossing must equal the BL time from Carter's t-equation plus $s\,T(r)$. The flare test suite verifies this to $10^{-7}\,M$ for three rays of the GRAVITY flare geometry. Building that test taught a lesson about the reference. A fresh Carter integration without the thesis' constraint projection, which re-imposes $p_r^2 = R(r)$ and $p_\theta^2 = \Theta(\theta)$ after every step, agreed with the Kerr–Schild time only to 10^{-4} at this near-face-on inclination. It converged to the Kerr–Schild value only as its tolerance was tightened, whereas the Kerr–Schild result was unchanged to twelve digits at every tolerance. With the projection restored, the two agree to 10^{-10}.

3.4 The live kernel

The GPU kernel is a single-precision port of the twin:

- classical fourth-order Runge–Kutta with steps of a few per cent of r, finer inside the photon region and near the disk plane;
- bisection of each equatorial crossing on the step fraction;
- capture at $r < r_+$;
- for escaping rays, the asymptotic direction from the coordinate velocity, so the background map has no parallax.

The seam heuristics of the BL kernel are gone, and the axis column renders like any other. Three quantitative comparisons with fp64 references underwrite it:

- the colour comparison of §2.3: 0.26 % in Doppler contrast;
- the moving-observer shadow edges of Chapter 4: 0.72 pixel;
- the flare images of Chapter 5: centroids within 0.27 pixel, total flux within 0.6 %.

Chapter 4

Whose Eyes? Observers in Motion

A black-hole image depends on who takes it. Two observers passing the same point at different velocities see the same light arrive from different directions (aberration), at different frequencies (Doppler shift) and with different brightness (beaming). Their clocks also run at different rates relative to a distant observer's. The thesis' camera was always a static observer far away, for whom none of this matters. This chapter gives the camera a 4-velocity.

4.1 Observers and their frames

A camera is an observer with 4-velocity $u = e_0$ and an orthonormal spatial triad e_i. A pixel looking along the unit direction n receives the photon with $k \propto e_0 - n^i e_i$, which the tracer follows backward (thesis Appendix on image handedness). The reference observer at each point is the zero-angular-momentum observer (ZAMO) of Bardeen et al. (1972),

$$e_{\hat{t}} = \frac{1}{\alpha}\left(\partial_t + \omega\,\partial_\phi\right), \qquad \alpha = \sqrt{-g_{tt} + g_{t\phi}^2/g_{\phi\phi}}, \quad \omega = -\frac{g_{t\phi}}{g_{\phi\phi}}, \tag{4.1}$$

with the spatial legs along the BL coordinate directions, normalised. An observer moving with velocity $\vec{v}$ relative to the ZAMO has the Lorentz-boosted tetrad

$$e'_0 = \gamma\,(e_{\hat{t}} + v^j e_{\hat{j}}), \qquad e'_i = e_{\hat{i}} + \frac{\gamma - 1}{v^2}\,v_i\,(v^j e_{\hat{j}}) + \gamma\,v_i\,e_{\hat{t}}, \tag{4.2}$$

orthonormal by construction. Aberration, Doppler shift and beaming of everything the camera sees then follow from tracing the received photons. Nothing is added by hand. The clock rate against far-away time is $d\tau/dt = 1/e'^t_0$.

The simulator offers five observers (`sgr_physics/observers.hpp`, shared by the renderer and the validation tools):

ZAMO: $\vec{v} = 0$, the frame-dragged observer.

Orbiting: circulating at the prograde Keplerian rate $\Omega = 1/(r^{3/2} + a)$, so $v_{\hat{\phi}} = (\Omega - \omega)\sqrt{g_{\phi\phi}}/\alpha$. This is a geodesic only on the equator.

Free fall: falling from rest at infinity with $E = 1$, $L_z = Q = 0$ — the "rain" frame of Hamilton & Lisle (2008) — with $v_{\hat{r}} = -\sqrt{2Mr(r^2 + a^2)/A}$. For this observer $\gamma = 1/\alpha$ exactly and there is no polar motion.

Hovering: at rest relative to the distant stars ($u \propto \partial_t$), so $v_{\hat{\phi}} = -\omega\sqrt{g_{\phi\phi}}/\alpha$. This observer needs a rocket, and becomes impossible ($v \to c$) at the ergosphere $r_E(\theta) = M + \sqrt{M^2 - a^2\cos^2\theta}$.

Riding an S-star: the star's coordinate velocity from its full-GR orbit, the same orbit model the thesis fits to the data (S2 passes pericentre at about 2.5 per cent of light speed, GRAVITY Collaboration, 2018).

The test suite (`test_observers.cpp`) checks these closed forms:

- every tetrad is orthonormal to 10^{-12};
- the orbiting observer's u^t, E and L are those of the BPT72 circular orbit;
- the free-fall observer has $u_t = -1$ and $u_\phi = 0$ at every (r, θ);
- the hovering observer's u^t is $1/\sqrt{-g_{tt}}$;
- hovering is valid at $1.001\,r_E$ and invalid at $0.999\,r_E$.

4.2 Clocks and speeds

Table 4.1 lists each observer's speed relative to the local ZAMO and its clock rate at three radii on the equator. The Schwarzschild entries reproduce textbook values: at the ISCO ($r = 6M$) the circular orbit moves at $0.5000\,c$ and its clock runs at $d\tau/dt = 0.7071 = 1/\sqrt{2}$, and the free-falling clock runs at $\alpha^2 = 1 - 2M/r$. For Sgr A*'s spin the hovering observer at $6M$ must already move at $0.0628\,c$ against the frame dragging just to stay still. The free-falling observer reaches $0.99989\,c$ relative to the ZAMO at $1.001\,r_+$ ($r_+ = 1.3412\,M$).

Table 4.1: Speed v relative to the local ZAMO [c] and clock rate $d\tau/dt$ (observer proper time per unit far-away time) of the four standard observers on the equator (`observer_tool`).

		ZAMO		orbiting		free fall		hovering	
a_*	r/M	v	$d\tau/dt$	v	$d\tau/dt$	v	$d\tau/dt$	v	$d\tau/dt$
0	60	0.000	0.9832	0.131	0.9747	0.183	0.9667	0.000	0.9832
0	15	0.000	0.9309	0.277	0.8944	0.365	0.8667	0.000	0.9309
0	6	0.000	0.8165	0.500	0.7071	0.577	0.6667	0.000	0.8165
0.94	60	0.000	0.9832	0.131	0.9748	0.183	0.9667	0.001	0.9832
0.94	15	0.000	0.9310	0.265	0.8978	0.365	0.8667	0.009	0.9309
0.94	6	0.000	0.8181	0.414	0.7448	0.575	0.6693	0.063	0.8165

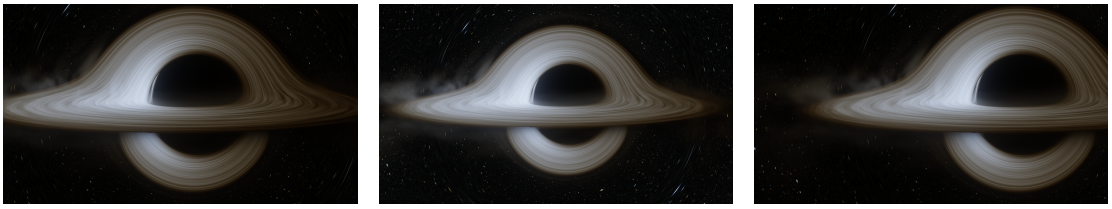


Figure 4.1: One place, three observers: the close-orbit scene seen by a ZAMO (left), by an observer falling from rest at infinity (centre) and by one orbiting at the Keplerian rate (right). Only the camera's tetrad differs.

4.3 What they see

Figure 4.1 shows the same place — the camera position of the close-orbit scene (Table 2.1) — through three observers' eyes. The free-falling observer sees the shadow and disk shrink toward the direction of motion and brighten and blue-shift: light from ahead is aberrated forward. The orbiting observer sees the whole scene displaced toward its direction of motion, brighter ahead and dimmer behind.

The shadow makes these effects quantitative because its edge has a closed form in the Schwarzschild case. A static observer at radius r sees a circular shadow of angular radius $\sin\psi = 3\sqrt{3}\,M\sqrt{1-2M/r}/r$ (Synge, 1966), which is $27.69°$ at $r = 10\,M$. A moving observer sees every direction aberrated, $\cos\chi' = (\cos\chi + v)/(1 + v\cos\chi)$ for the angle χ to its velocity. Bisecting the shadow edges on the fp64 Kerr–Schild tracer with each observer's tetrad (`observer_tool`) reproduces this prediction to 2×10^{-12} in the tangent of the edge angle:

- the free-falling observer ($v = 0.4472$) sees a shadow of radius $17.32°$;

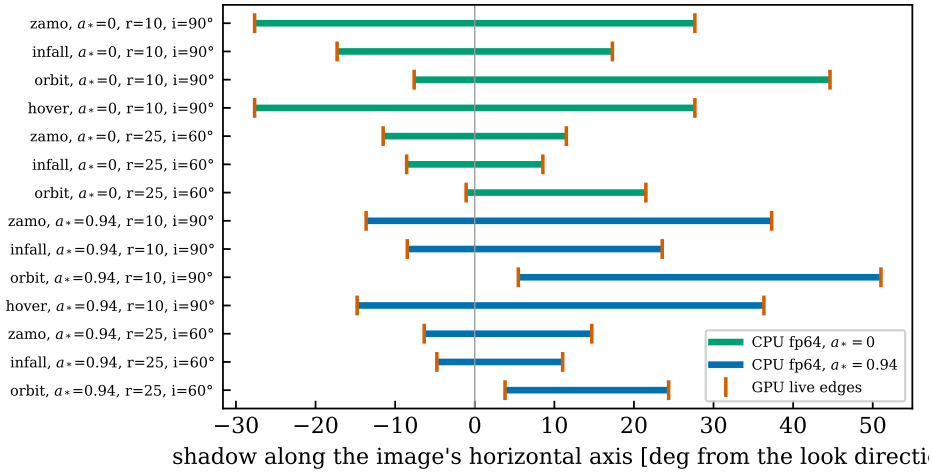


Figure 4.2: The shadow along the image's horizontal axis for each observer and camera position (bars: fp64 Kerr–Schild bisection; ticks: the live GPU renderer). At $a_* = 0$ the bars equal Synge's shadow aberrated by the observer's velocity.

- the orbiting observer ($v = 0.3536$) sees its edges at 7.65° on the trailing side and 44.65° on the leading side.

At $a_* = 0.94$ the ZAMO's shadow is already lopsided, with edges at 13.66° (prograde) and 37.32°. The orbiting observer's aberration carries the whole shadow past the centre of view, to between 5.47° and 51.04° on the leading side (Fig. 4.2).

4.4 The sky they see

The background shifts too. For a backward ray normalised so the camera measures unit frequency, the conserved p_t gives the ratio of received to emitted frequency for light from infinity, $g_{\mathrm{sky}} = 1/p_t$. Deep in the potential the sky is blueshifted by $1/\alpha$ for a ZAMO, more ahead of a falling observer and less behind it. The renderer applies this shift to the Milky Way map using the 6500 K blackbody approximation of §2.6. The effect is weak for these exterior observers, but it dominates the view near the horizon (Chapter 8).

4.5 Frame dragging made visible

Inside the ergosphere nothing can remain at rest relative to the stars, and light is dragged too. Every captured ray crosses the ergosurface, so tinting those rays

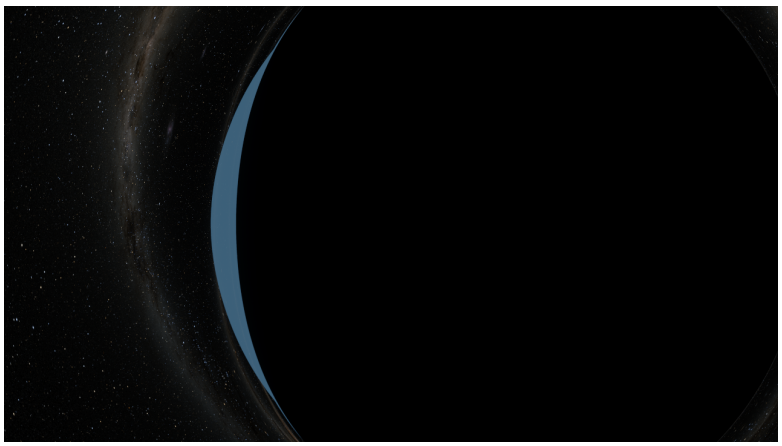


Figure 4.3: Light that passed through the ergoregion and escaped (blue overlay) around Sgr A* seen nearly edge-on ($i = 88°$, $r = 30\,M$, no disk; lensed Milky Way behind). Captured light is not marked. The crescent appears only on the prograde side, where photon orbits dip inside the ergosphere.

would only colour the shadow. The overlay instead marks the light that went *inside* r_E and still escaped. That set is not symmetric. At $a_* = 0.94$ the prograde equatorial photon orbit, at $1.43\,M$, lies inside the ergosphere ($2\,M$ on the equator), while the retrograde one, at $3.95\,M$, lies outside it. Only light passing on the prograde side can visit the ergoregion and return, and it does so in a thin crescent hugging that side of the shadow (Fig. 4.3). The crescent is a direct picture of frame dragging.

4.6 Validating the live observers

The live renderer receives the boosted tetrad (4.2), computed in double precision by the same library function the CPU tools use and expressed in Kerr–Schild components. `compare_observers.py` renders a one-pixel-high strip through the image centre against a uniform sky, with no disk, so captured rays are exact zeros. It then compares the GPU shadow edges with the fp64 edges for 14 observer and spin configurations. The largest difference is 0.72 pixel (one pixel is 0.00140 in the tangent of the angle).

That agreement came only after the comparison found a defect. The live kernel gave each ray at most 1600 steps and counted a ray that exhausted them as captured. Near the high-spin prograde photon orbit ($r = 1.43\,M$ at $a_* = 0.94$) the orbit is only weakly unstable. A ray one pixel from the shadow edge winds

several times at about a thousand steps per orbit, ran out of budget, and was wrongly darkened. Edge-on, this widened the prograde side of the shadow by 16.8 pixels at the comparison's resolution. The budget is now 6000 steps. Only rays at the photon ring use the extra steps, and the old budget can be re-run on demand to reproduce the record.

Chapter 5

Orbiting Hot Spots: the GRAVITY Flares

Several times a day the near-infrared emission of Sgr A* flares, rising within minutes to many times its quiescent level (Genzel et al., 2010). In 2018 the GRAVITY interferometer resolved the motion of three bright flares (GRAVITY Collaboration et al., 2018). Their emission centroids traced clockwise loops on the sky, with the polarisation angle rotating in step. On July 22 the centroid moved by about $120\,\mu$as in about 30 minutes, roughly 30 per cent of the speed of light. GRAVITY's relativistic modelling located the emitter on a near face-on circular orbit just outside the innermost stable orbit, with best fits near $R \approx 7 \pm 0.5\,R_g$, $i \approx 160^\circ \pm 10^\circ$ and a period $P = 40 \pm 8$ minutes. The other two flares traced loops of total amplitude $\sim$110–140 μas. A dedicated follow-up (GRAVITY Collaboration et al., 2020) found all three flares consistent with a compact hot spot at $R \approx 9\,R_g$ and $i \approx 140^\circ$, and noted a tension: the observed motion on the sky is systematically *larger* than geodesic hot-spot models predict for the observed period.

This chapter builds that model on our Sgr A* spacetime, traces it with full slow-light ray tracing, and compares its observables with the published ones. Every orbit and inclination below is taken from those papers; nothing is tuned.

5.1 The model

The emitter is the one GRAVITY used (GRAVITY Collaboration et al., 2020, following Broderick & Loeb 2006; Hamaus et al. 2009): a Gaussian blob of emissivity $\varepsilon \propto \exp(-d^2/2s^2)$, size $s = 0.5\,M$ (their fiducial value), whose centre follows the

prograde circular equatorial geodesic of BL radius R, $\Omega = 1/(R^{3/2} + a)$. The blob moves rigidly with the centre's 4-velocity $u = u^t(\partial_t + \Omega\,\partial_\phi)$, and d is the comoving distance $d^2 = (g_{\mu\nu} + u_\mu u_\nu)\,\Delta x^\mu \Delta x^\nu$ at fixed coordinate time. The emission is optically thin, with $j_\nu \propto \nu^{-\alpha}$. Along a backward ray normalised so the camera measures unit frequency, conservation of I_ν/ν^3 gives

$$I_{\rm obs} = \int g^{2+\alpha}\,\varepsilon\,d\lambda, \qquad g = \frac{1}{u^t\,(p_t + \Omega L_z)}, \tag{5.1}$$

where g is one number per ray for a rigidly rotating emitter. With $\alpha = 0$ this is exactly the GRAVITY 2020 prescription, $F = \int D^2\varepsilon\,ds$. The value $\alpha = 0.6$, one of the spectral indices GRAVITY used to illustrate beaming (GRAVITY Collaboration et al., 2018, their Appendix C), measures the sensitivity to that choice.

Slow light. Each ray carries its Kerr–Schild time (Chapter 3). Light reaching the camera at observing time $t_{\rm obs}$ left a point on the ray at $t_{\rm obs} + \Delta t$, when the spot was where it was then. The simulator does not freeze the source while the light travels.

Ray table. The spacetime is stationary, so each pixel's ray is traced once (fp64, adaptive Dormand–Prince) from a camera at $1000\,M$ over a field of $\pm 14\,M$ with 256^2 pixels. The tracer stores every sample within $6s$ of the orbit, at a spacing of at most $s/5$, together with its Kerr–Schild time and its image order: 0 for the direct image, 1 for the first image lensed around the hole. Any observing time and emission model is then a sum over stored samples. Along all rays $|H|/E^2 \le 5 \times 10^{-10}$ and the relative drift of L_z is $\le 1 \times 10^{-10}$.

Tests (`test_hotspot.cpp`) cover four properties:

- the orbit closes after exactly $2\pi(R^{3/2} + a)$;
- the comoving distance co-rotates with the spot;
- the Kerr–Schild light-travel time matches Carter's t-equation (Chapter 3);
- cameras at i and $180^\circ - i$ record identical light curves and mirror-image centroid tracks, as the equatorial symmetry of Kerr requires.

Numerical noise. The lensed images are thin arcs that no affordable grid resolves. As they move across pixels they add a pixel-scale jitter that shrinks as $1/n$. All observables below use the first twelve harmonics of the one-orbit series, which removes 0.22 % (rms) from the light curve and $0.16\,\mu$as from the centroid

Table 5.1: Simulated observables of an orbiting hot spot around Sgr A* ($a_* = 0.94$; $s = 0.5\,M$), from `flare_tool`. Loop: the largest separation of two centroid positions; in brackets the geometric orbit diameter. Chord: the largest centroid displacement within 30 minutes and the corresponding speed. Contrast: maximum over minimum flux for $\alpha = 0$ ($\alpha = 0.6$). Lensed: mean flux fraction in images of order $n \geq 1$. Echo: delay of the $n = 1$ image of a flash behind the direct one, over eight orbital phases.

	GRAVITY 2018	GRAVITY 2020	$P = 40\,\mathrm{min}$
R [M], i [deg]	7.00, 160	9.00, 140	6.64, 160
period [min]	43.1	61.9	40.0
loop [μas]	74.4 (71.7)	96.6 (92.2)	70.5 (68.0)
30-min chord [μas] (c)	64 (0.15)	97 (0.22)	54 (0.12)
contrast	2.23 (2.60)	3.92 (5.11)	2.27 (2.65)
lensed flux [%]	6.5	4.1	7.1
echo delay [min]	5.9–9.1	4.7–16.6	5.8–8.9

track. At half the resolution (128^2 pixels) the loop size changes by 1.7 %, the flux contrast by 3.0 % and the echo delay by 0.6 %. These are the numerical uncertainties of the values below.

5.2 Comparison with GRAVITY

Table 5.1 and Figure 5.1 give the results for three orbits:

- GRAVITY's 2018 best fit;
- their 2020 global fit;
- the radius whose prograde period, for our spin and the GRAVITY 2022 mass, equals the 2018 best-fit period of 40 minutes ($R = 6.64\,M$).

Period and sense. On GRAVITY's 2018 orbit the spot circles Sgr A* in 43.1 minutes, inside the measured 40 ± 8. Seen from $i > 90°$ its motion is clockwise on the sky, as observed. The 2020 orbit, at $9\,M$, takes 61.9 minutes: within the 33–65-minute range GRAVITY quoted as consistent with all three flares, but longer than the July 22 best fit. The two radii bracket the problem.

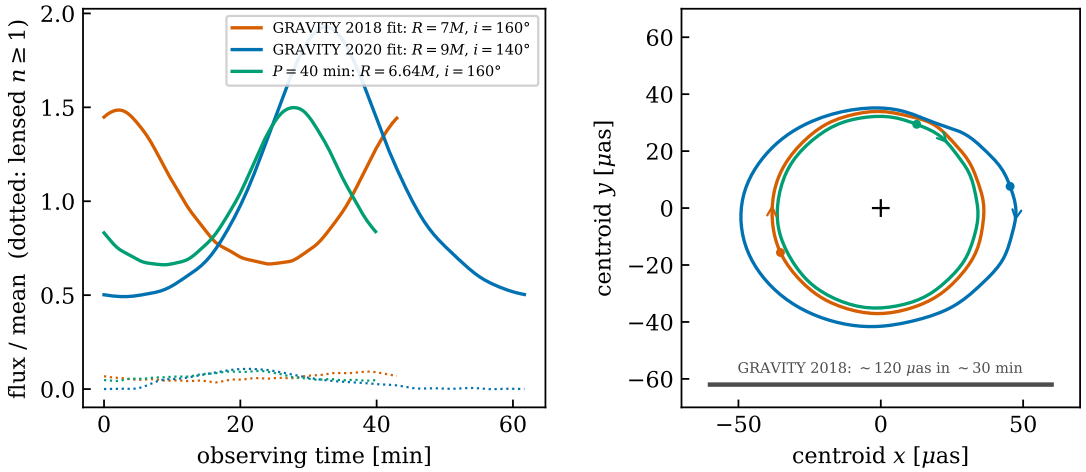


Figure 5.1: Hot spots on the GRAVITY orbits around Sgr A*. Left: light curves over one orbit, normalised to the mean (dotted: the part carried by lensed images). Right: the flux-weighted centroid tracks on the sky (dot: start; arrows: sense of motion; cross: the black hole). The grey bar shows the centroid motion GRAVITY measured in the July 2018 flare, about 120 μas in about 30 minutes (GRAVITY Collaboration et al., 2018).

Loop size and speed. Lensing enlarges the apparent orbit a little (as GRAVITY's models also show; GRAVITY Collaboration et al., 2018): the 2018 orbit's centroid loop measures 74.4 μas against a geometric diameter of 71.7 μas. Within any 30 minutes, however, the centroid moves at most 64 μas (0.15 c), about half the ~120 μas (~0.3 c) GRAVITY measured in the same interval. Moving the spot out to 9 M at $i = 140°$ brings the 30-minute chord to 97 μas, but at a period of 61.9 minutes. No circular geodesic reconciles both. This reproduces, with an independent code and our own spacetime, the tension of GRAVITY Collaboration et al. (2020): the observed motion is faster and larger than a hot spot on a geodesic can be for the observed period. They discuss two partial explanations, neither modelled here: astrometric scatter biases a measured radius upward by $\sqrt{1 + \sigma^2/R^2}$, and the spot may move toward the observer.

Flux contrast. On the 2018 orbit the light curve varies by a factor 2.23 for $\alpha = 0$ (2.60 for $\alpha = 0.6$). This is the combined effect of Doppler beaming, lensing and the changing path through the blob. GRAVITY's combined orbit constraints required a Doppler contrast below about 2.5 (GRAVITY Collaboration et al., 2018, their Fig. B.3): the 2018 orbit meets that with $\alpha = 0$ and only marginally

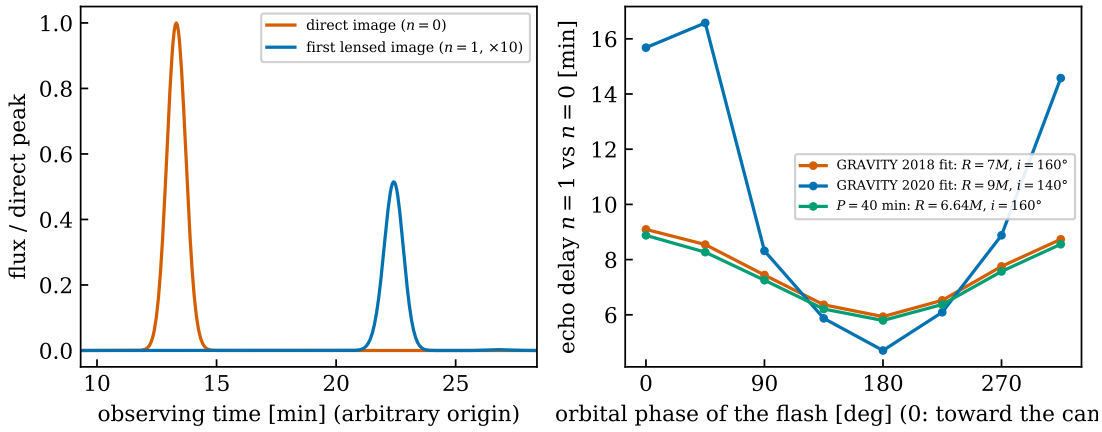


Figure 5.2: Photon-ring echoes on the GRAVITY 2018 orbit. Left: a brief flash ($\tau = 1\,M$) seen in the direct image and, delayed, in the first lensed image (scaled up tenfold). Right: the echo delay versus the orbital phase of the flash for the three orbits of Table 5.1.

with $\alpha = 0.6$. The 2020 geometry, more inclined, varies by 3.92. Its intrinsic heating and cooling would have to mask a beaming signal of that size. The loop centres lie 2.1 μas (2018) and 4.9 μas (2020) from the black hole, well inside the $\pm 50\,\mu$as within which GRAVITY found them consistent with the mass centroid.

5.3 Photon-ring echoes

Light from the spot also reaches the camera along a second path, around the hole, as the thin lensed image in Fig. 5.3. That image carries 6.5 per cent of the flux on average on the 2018 orbit. Because the second path is longer, a sudden change in the source appears twice. Flashing the spot briefly ($\tau = 1\,M$) at eight points of its orbit and following the direct and the first lensed image separately (Fig. 5.2) gives echo delays of 16.8–25.8 M, that is 5.9–9.1 minutes (mean 7.5). The echo peaks at 2.8–11.4 per cent of the direct flash. The delay depends on the phase because the extra path depends on where the spot is relative to the line of sight; at the more inclined 2020 geometry it spans 4.7–16.6 minutes. These are the time-domain counterpart of the thesis' photon-ring analysis: a delay set by the mass and geometry alone, measurable in principle without resolving the ring.

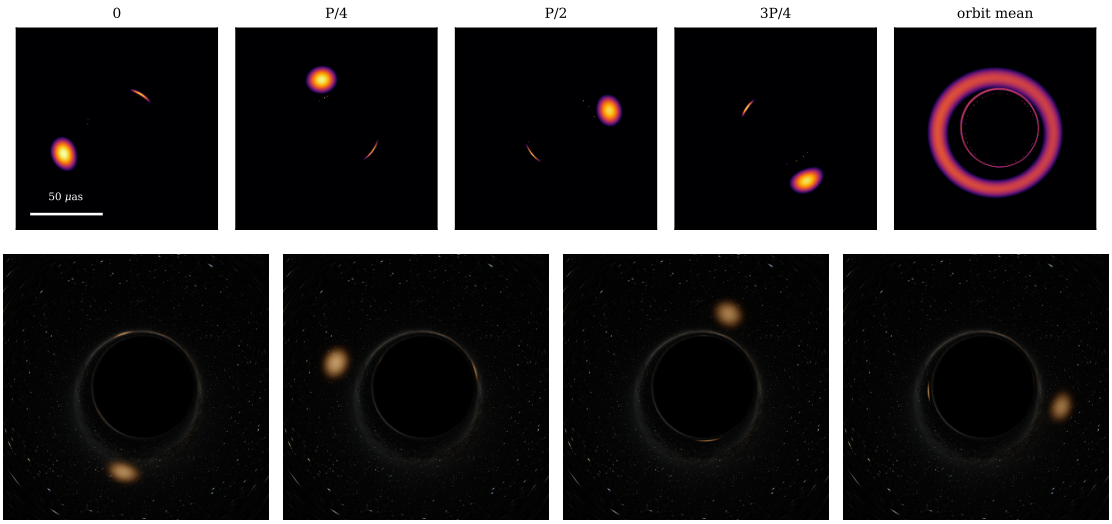


Figure 5.3: The GRAVITY 2018 hot spot around Sgr A* ($R = 7\,M$, $i = 160°$). Top: fp64 images (logarithmic stretch, four decades) at quarter orbits and averaged over one orbit; the thin arcs near the shadow edge are the lensed images. Bottom: the live renderer at the same phases from $45\,M$, with the lensed Milky Way behind and the spot in false colour.

5.4 The live view

The live renderer traces the same emitter: the orbit constants and the comoving metric are computed by the same CPU class and handed to the shader, and each ray integrates its own Kerr–Schild time. Against the fp64 images at four orbital phases (`compare_flare.py`) the GPU total flux agrees to 0.6 %, the centroid to 0.27 pixel, and the image correlation is at least 0.992. Figure 5.3 shows the fp64 images. In the interactive renderer and the movie of one orbit (`scripts/make_flare_movie.py`) the spot is drawn in false colour, because 2.2 μm synchrotron light has no visible hue.

Chapter 6

The Hot Accretion Flow at 230 GHz

Chapter 2 showed that Sgr A* cannot be fed by a thin disk: at the measured accretion rate one would outshine it a hundredfold. Its accretion flow is instead hot, geometrically thick and optically thin over most of the spectrum, a radiatively inefficient accretion flow (RIAF), and its emission peaks at the sub-millimetre wavelengths the Event Horizon Telescope observes. This chapter builds such a flow, traces its 230 GHz synchrotron emission with radiative transfer along the Kerr–Schild rays of Chapter 3, normalises it to the measured flux density, and compares the result with the EHT's total-intensity (Event Horizon Telescope Collaboration, 2022) and polarised (Event Horizon Telescope Collaboration et al., 2024a,b) images of Sgr A*.

6.1 The model

Plasma. We adopt the semi-analytic flow that Broderick and collaborators used to interpret the first millimetre-VLBI data of Sgr A* (Broderick & Loeb, 2006; Broderick et al., 2011), after Yuan et al. (2003). With cylindrical radius ρ and height z (from the Boyer–Lindquist r and θ), the thermal electrons have

$$n_e = n_0 \left(\frac{\rho}{M}\right)^{-1.1} e^{-z^2/2\rho^2}, \qquad T_e = T_0 \left(\frac{r}{M}\right)^{-0.84}, \tag{6.1}$$

with $T_0 = 1.5 \times 10^{11}\,\mathrm{K}$, the value of Broderick & Loeb (2006) for their high-spin model. The field is toroidal and in partial equipartition with the ions, $B^2/8\pi = \beta^{-1} n_e m_p c^2 r_S/12r$ with $\beta = 10$ (Broderick et al., 2011). The gas circulates at the Keplerian rate of its cylindrical radius outside the innermost stable orbit

and plunges inside it. On the equator it follows the plunging geodesic with the ISCO's energy and angular momentum. Off the plane every point takes that plunge's radial and angular velocity at the same r, normalised locally. Only thermal electrons are modelled. Broderick et al. (2011) attribute roughly 30 per cent of the 1.3 mm flux in their models to a non-thermal population, which we omit.

Emission and transfer. The thermal synchrotron emissivity is the fit of Leung et al. (2011), as implemented in grmonty (Dolence et al., 2009):

$$\begin{aligned} j_\nu &= \frac{\sqrt{2}\,\pi e^2 n_e \nu_s}{3cK_2(1/\Theta_e)}\left(X^{1/2}+2^{11/12}X^{1/6}\right)^2 e^{-X^{1/3}}, \\ X &= \frac{\nu}{\nu_s}, \qquad \nu_s = \frac{2}{9}\,\frac{eB}{2\pi m_e c}\,\Theta_e^2 \sin\theta_B, \end{aligned} \tag{6.2}$$

where $\Theta_e = kT_e/m_ec^2$ and θ_B is the angle between the photon and the field in the fluid frame. Absorption follows from Kirchhoff's law, $\alpha_\nu = j_\nu/B_\nu(T_e)$. Along a backward ray normalised so the camera measures unit frequency, the fluid sees $\varepsilon = p\cdot u$, so $\nu_{\rm f} = \varepsilon\,\nu_{\rm obs}$, and the proper length per affine step is $\varepsilon\,\Delta\lambda$. For a thermal source the conserved I_ν/ν^3 gives the composited solution

$$I_{\rm obs} = \sum_k \frac{B_{\nu_{\rm f}}(T_e)}{\varepsilon^3}\left(1-e^{-\Delta\tau_k}\right)e^{-\tau_k}, \qquad \Delta\tau_k = \alpha_{\nu_{\rm f}}\,\varepsilon\,\Delta\lambda\,r_g, \tag{6.3}$$

accumulated from the camera outward. Faraday rotation and conversion are not modelled.

Normalisation and tests. The density normalisation is the one free parameter. It is fitted, by bisection on full images, so that the 230 GHz flux density equals the EHT benchmark of 2.4 Jy (Event Horizon Telescope Collaboration, 2022). This gives $n_0 = 2.7\times10^7\,\mathrm{cm^{-3}}$. At the ISCO the plasma then has $n_e = 1.2\times10^7\,\mathrm{cm^{-3}}$, $T_e = 8.3\times10^{10}\,\mathrm{K}$ and $B = 62\,\mathrm{G}$. Images span $\pm12\,M$ with 160^2 pixels from a camera at $1000\,M$ (at 120^2 pixels the ring diameter below changes by 0.7 %), and $|H| \le 3\times10^{-9}$ along every ray. The test suite (`test_riaf.cpp`) checks four properties:

- the flow is a unit timelike vector and the field a unit vector orthogonal to it, inside and outside the ISCO;
- (6.3) returns $B_\nu(T)$ for an optically thick slab and $\sum j\,dl$ for a thin one;

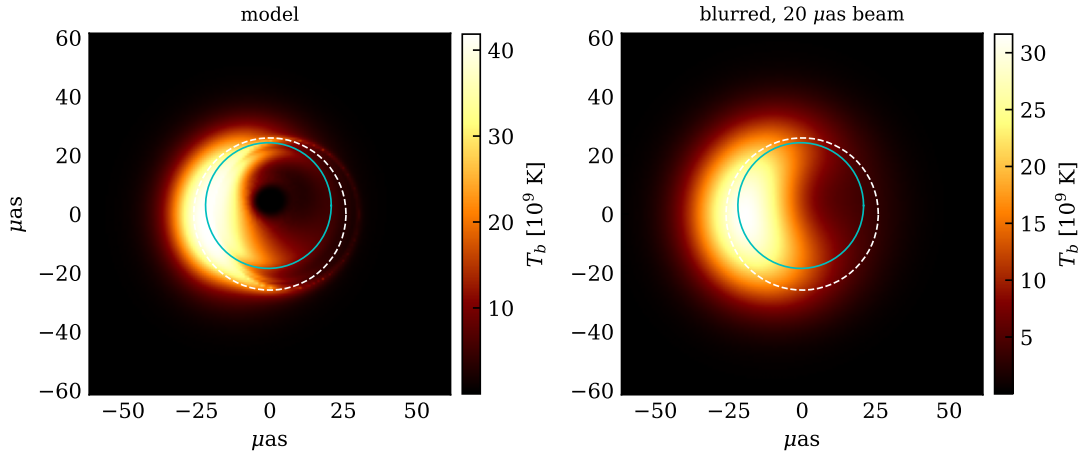


Figure 6.1: Sgr A* at 230 GHz in the RIAF model ($a_* = 0.94$, $i = 30°$, n_0 fitted to 2.4 Jy), as brightness temperature. Left: the model image. Right: blurred with the EHT's 20 μas restoring beam. Dashed white: the EHT ring diameter, 51.8 μas (Event Horizon Telescope Collaboration, 2022); cyan: the ring measured on the model image.

- j_ν is linear in n_e, vanishes along the field and falls off exponentially above ν_s;
- images seen from either side of the equatorial plane are exact mirror images.

6.2 The image and the EHT ring

Figure 6.1 shows the result at the EHT viewing geometry ($a_* = 0.94$, $i = 30°$). Most of the light comes from a crescent on the approaching side. It is the direct image of the hot, Doppler-boosted inner flow and lies *inside* the photon ring, which is visible as a thin circle at the critical curve. The inner flow's horizon silhouette, the "inner shadow" (Chael et al., 2021), appears as the dark spot at the centre. The peak brightness temperature is 4.2×10^{10} K, or 3.2×10^{10} K after blurring with the 20 μas restoring beam of the EHT's CLEAN images.

The ring that the brightness peaks trace on the model image, measured as the EHT's REx procedure does it, has a diameter of 42.8 μas (the centre minimises the spread of peak radii over 180 position angles). The EHT measured 51.8 ± 2.3 μas. After the 20 μas beam the model's crescent is no longer resolved as a ring: its width exceeds its diameter, $W/d = 1.15$. The comparison that does not depend on resolution is the visibility amplitude: blurring multiplies the visibilities by a

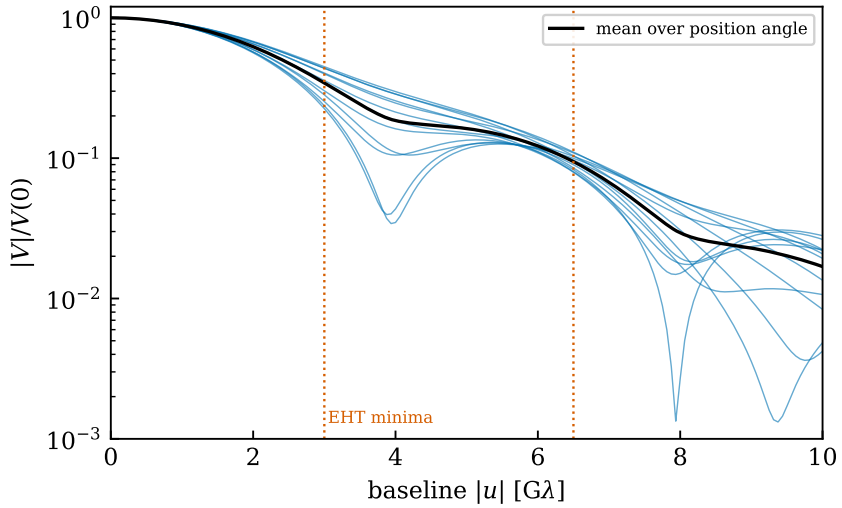


Figure 6.2: Visibility amplitudes of the model image of Fig. 6.1 along twelve baseline position angles (blue) and averaged over all (black). Dotted: the minima of the EHT data (Event Horizon Telescope Collaboration, 2022). A beam does not move these minima.

Gaussian and leaves their minima where they are. The EHT found its first and second minima at $|u| \approx 3.0$ and 6.5 Gλ, like a thin ring of about 54 μas (Event Horizon Telescope Collaboration, 2022). Along the position angles where our model shows a minimum (26 of 36; the lopsided crescent fills in the others), the first lies at 3.90 to 9.95 Gλ. The shortest is equivalent to a thin ring of 40.5 μas (Fig. 6.2).

This model is therefore too compact for Sgr A*. Its 230 GHz light is dominated by the inner flow rather than by the photon ring, and its ring is 17 per cent smaller than the one observed. This is the caveat the thesis attached to its calibration (the measured ring tracks the critical curve only as well as the emitting plasma sits on it), made quantitative for one physical model. The sensitivity runs of Table 6.1 show where the model would have to change. Halving the electron temperature moves the emission outward and makes the image nearly symmetric: the first minimum appears along every position angle, at 3.20 Gλ (a thin ring of 49 μas), and the brightness asymmetry drops to 0.08. A nearly face-on view ($i = 10°$) gives a ring of 49.2 μas. The field strength barely matters once n_0 is refitted.

The live renderer offers the same flow as its 230 GHz band. With identical parameters and pixel grid it reproduces the fp64 image to 0.2 % in total flux, with a correlation of 0.9999 and centroids within 0.03 pixel (`compare_riaf.py`).

Table 6.1: The RIAF at 230 GHz: fitted density normalisation, ring diameter on the model image, first visibility minimum (shortest over position angles; in brackets the equivalent thin-ring diameter) and brightness asymmetry after the 20 μas beam, for the reference model and variations (each refitted to 2.4 Jy). EHT values from Event Horizon Telescope Collaboration (2022).

Case	n_0 [cm^{-3}]	d [μas]	first null [Gλ] (μas)	A
reference (T_0, $\beta = 10$, $i = 30°$)	2.7×10^7	42.8	3.90 (40)	0.33
$T_0/2$	1.7×10^8	36.1	3.20 (49)	0.08
$2T_0$	9.2×10^6	47.0	4.45 (35)	0.48
$\beta = 3$	1.4×10^7	43.3	3.85 (41)	0.35
$\beta = 30$	5.1×10^7	42.5	3.95 (40)	0.32
$i = 10°$	3.1×10^7	49.2	3.55 (44)	0.16
$i = 60°$	2.3×10^7	27.2	4.25 (37)	0.37
EHT 2022 (Paper I)	—	51.8 ± 2.3	≈ 3.0 (≈ 54)	0.04–0.3

Figure 6.3 shows one camera in the three bands the simulator now renders.

6.3 Polarisation

Synchrotron light is linearly polarised with its electric vector perpendicular to both the photon direction and the magnetic field in the emitting frame, $f^\mu \propto \varepsilon^{\mu\nu\rho\sigma} u_\nu k_\rho b_\sigma$. Along a Kerr null geodesic the polarisation vector is parallel- transported, and the Walker & Penrose (1970) constant

$$\kappa = (A - iB)(r - ia\cos\theta), \quad A = (k^t f^r - k^r f^t) + a\sin^2\theta\,(k^r f^\phi - k^\phi f^r), \quad (6.4)$$

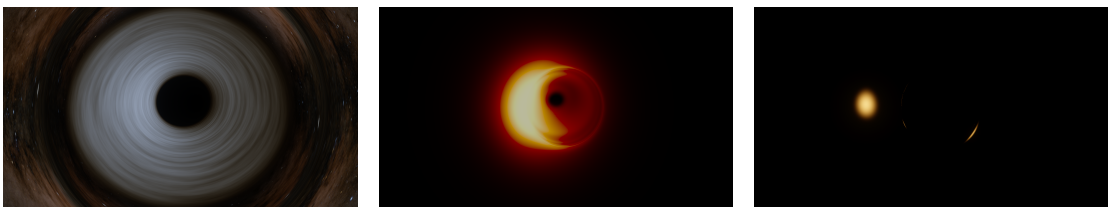


Figure 6.3: One camera ($i = 30°$, 60 M), three bands. Left: visible light, with the hypothetical thin disk of Chapter 2 and the lensed Milky Way. Centre: 230 GHz, the hot accretion flow of this chapter (brightness temperature, false colour). Right: 2.2 μm, a GRAVITY hot spot (Chapter 5; false colour). Only the 230 GHz view shows Sgr A*'s real quiescent flow.

$$B = \left[(r^2 + a^2)(k^\phi f^\theta - k^\theta f^\phi) - a\,(k^t f^\theta - k^\theta f^t)\right] \sin\theta$$

(Connors et al., 1980; Himwich et al., 2020) carries its orientation to the camera without integrating the transport. At the camera the electric-vector position angle (EVPA) is the angle χ in the image plane for which $\kappa(\cos\chi\, e_1 + \sin\chi\, e_2)$ is a real multiple of the emission point's κ, with $e_{1,2}$ the camera's image axes made orthogonal to the pixel direction.

The tests (`test_polarisation.cpp`) cover three properties:

- κ is conserved to 10^{-8} when the polarisation vector is parallel-transported explicitly with the Christoffel symbols along a Kerr geodesic;
- a face-on, non-rotating flow with a toroidal field produces a purely radial EVPA pattern;
- the views from either side of the equator give complex-conjugate patterns.

Each emitting element contributes Stokes $Q + iU = p_0\, dI\, e^{2i\chi}$ with a constant intrinsic polarisation fraction $p_0 = 0.7$, an assumption. Faraday rotation and conversion are omitted.

The EHT summarises the polarisation pattern of a ring by its azimuthal modes $\beta_m = \sum P e^{-im\phi} / \sum I$ (Palumbo et al., 2020). The phase $\angle\beta_2$ measures how the EVPAs twist around the ring: 0 for a radial pattern, $\pm 180°$ for a toroidal one. It is invariant under rotation of the image, so the unknown orientation of Sgr A*'s spin on the sky drops out. Its sign follows the sense in which the flow rotates on the sky. With p_0 an assumption, the comparable quantities are $\angle\beta_2$ and $|\beta_2|/|\beta_1|$, which do not depend on it.

For each field geometry we refit n_0 to 2.4 Jy and view the flow from $i = 30°$ (counter-clockwise on the sky) and 150° (clockwise, the sense of the GRAVITY flares). Table 6.2 and Fig. 6.4 give the results:

- *toroidal field* — the electric vectors point radially, $|\angle\beta_2| \approx 1°$. The EHT measured $\angle\beta_2$ far from zero (Event Horizon Telescope Collaboration et al., 2024a), so a purely toroidal field is excluded;
- *radial field* — a near-toroidal pattern, $\angle\beta_2 = -168°$ (counter-clockwise) or $+168°$ (clockwise);
- *vertical (poloidal) field* — a spiral, $\angle\beta_2 = +135°$ for counter-clockwise rotation, inside the EHT's as-observed range of 125° to 159°, and $-135°$ for

Table 6.2: Polarisation of the RIAF at 230 GHz for three field geometries, seen with counter-clockwise ($i = 30°$) and clockwise ($i = 150°$) rotation on the sky: phase of β_2 on the model and blurred images (degrees), and on the blurred image $|\beta_2|$, $|\beta_2|/|\beta_1|$, the resolved $\langle|m|\rangle$ and the net $|m_{\rm net}|$. $|\beta_2|$, $\langle|m|\rangle$ and $|m_{\rm net}|$ scale with the assumed intrinsic fraction $p_0 = 0.7$. EHT 90 per cent ranges from Event Horizon Telescope Collaboration et al. (2024a).

field	i [deg]	$\angle\beta_2$	$\angle\beta_2^{\rm blur}$	$\lvert\beta_2\rvert$	$\lvert\beta_2\rvert/\lvert\beta_1\rvert$	$\langle\lvert m\rvert\rangle$ [%]	$\lvert m_{\rm net}\rvert$ [%]
toroidal	30	+3	+1	0.43	1.5	47	6.8
toroidal	150	-3	-1	0.43	1.5	47	6.8
radial	30	-163	-168	0.43	2.7	44	5.4
radial	150	+163	+168	0.43	2.7	44	5.4
vertical	30	+133	+135	0.27	0.8	42	12.5
vertical	150	-133	-135	0.27	0.8	42	12.5
EHT 2024, observed	—	—	125 to 159	0.14–0.24	1.1–2.1	24–28	—
EHT 2024, derotated	—	—	−168 to −85				

clockwise rotation, inside the range of $-168°$ to $-85°$ the EHT obtains after removing a constant (external) Faraday rotation measure (Event Horizon Telescope Collaboration et al., 2024a). Its $|\beta_2|/|\beta_1| = 0.8$ compares with the EHT's 1.1–2.1.

Only the poloidal field reproduces the observed spiral. That agrees with the EHT's own interpretation, which favours flows with dynamically important magnetic fields (Event Horizon Telescope Collaboration et al., 2024b). Which sign of $\angle\beta_2$ applies depends on how much of the observed rotation measure is external to the emitting region, a question this model, which omits Faraday effects, cannot settle.

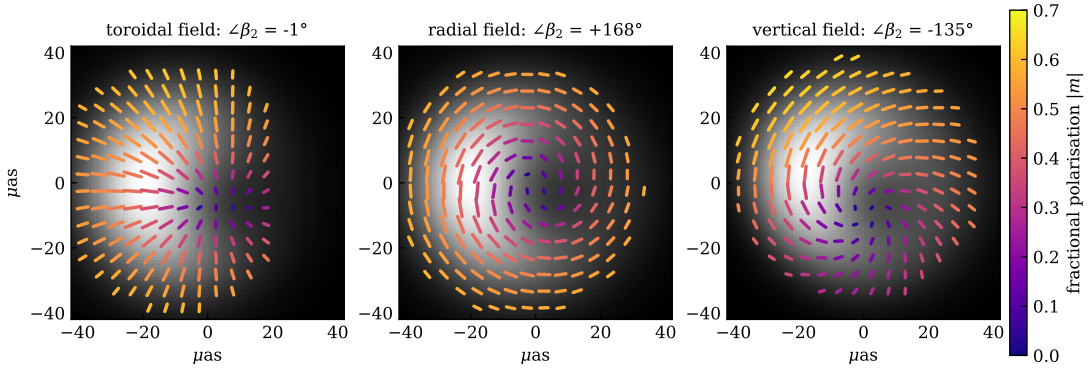


Figure 6.4: EVPA patterns of the 230 GHz RIAF for three magnetic-field geometries, with the flow rotating clockwise on the sky ($i = 150°$), blurred with the EHT beam. Grey: total intensity. Ticks: EVPA, length proportional to the polarised intensity, colour to the fractional polarisation. Only the vertical (poloidal) field gives the spiral seen by the EHT.

Chapter 7

Forecasts for Orbit Monitoring

The thesis measured the mass and distance of Sgr A* from the orbit of S2 and bounded the no-hair parameter ε_q with the Gillessen et al. (2017) data. This chapter asks what continued monitoring of S2 can add, and what could make it mislead. It uses the thesis' own forward model, the full GR orbit with Rømer delay and relativistic redshift, on synthetic data. Four questions are addressed:

1. how fast the no-hair constraint improves as orbits accumulate;
2. when S2 can reveal frame dragging;
3. how an extended mass around Sgr A* couples to the no-hair parameter;
4. how large a false no-hair signal an extended mass produces if it is left out of the fit.

7.1 Method

Truth and data. The true orbit is the thesis' S2 fit, with the reference-frame nuisances set to the centres of their priors. Noise-free synthetic data are made with the forward model at three kinds of epochs.

- The real Gillessen et al. (2017) epochs and errors: 145 astrometric and 44 radial-velocity points spread over 24.3 yr.
- A *GRAVITY-era* continuation from 2017.0, with 18 astrometric epochs per year at 65 μas per coordinate and 5 radial velocities per year at 12 km s^{-1}. These are the precision and cadence GRAVITY reports for 2016.7–2019.7: 54 interferometric positions with rms uncertainty $\approx 65\,\mu$as, and spectra with

$\approx 12\,\mathrm{km\,s^{-1}}$ uncertainty taken 3–6 times a year (GRAVITY Collaboration, 2020).

- An *existing-quality* continuation that repeats the adaptive-optics-era sampling of the Gillessen et al. (2017) data: 9.2 positions per year at 0.4 mas, and 2.7 radial velocities per year at $45\,\mathrm{km\,s^{-1}}$ (medians of those data).

Sampling is uniform in time, with no seasonal gaps, confusion or frame systematics, so the forecasts below are idealised.

Forecasts and biases. Forecast uncertainties come from the Fisher matrix, $F = J^{\mathrm{T}} W J$ plus the thesis' frame priors. Here J is the central-difference Jacobian of the forward model with respect to its 13 orbit and frame parameters, plus whichever of ε_q, a_* and the extended mass are free, and W holds the inverse data variances.

The finite-difference steps need care. Around a pericentre passage S2 moves fastest, along a sharply curved path, so anything that shifts the time of passage (the mass, distance, semi-major axis or ε_q) changes the predicted positions nonlinearly. With the thesis' steps, which were tuned for its imaging-era data, the Fisher matrix for the GRAVITY-era data sets therefore depended on the steps. Here those steps are four to eight times smaller. Halving or doubling any of them, together or one at a time, now moves no forecast quoted in this chapter by more than 2.3 per cent.

Three further checks support the method. On the Gillessen et al. (2017) epochs, the Fisher matrix gives $\sigma(\varepsilon_q) = 93\,500$, against 93 500 from the thesis' profile-likelihood fit to the real data (0.0 per cent apart), and $\sigma(M)$ agrees to 0.09 per cent. For the GRAVITY-era data sets, the Fisher $\sigma(\varepsilon_q)$ agrees with the curvature of full profile-likelihood fits within 8 per cent (below). Biases, finally, are not linearised at all. An effect that is present in the data but absent from the model is fitted with the thesis' own fitter: the 13-parameter Ceres fit for M and R_0, and the thesis' profile likelihood in ε_q (13-parameter fits at fixed ε_q). The profile is not optional. Ceres differentiates numerically with a step of 10^{-6} times the parameter, floored at $\sqrt{\epsilon_{\mathrm{mach}}} \approx 1.5 \times 10^{-8}$. At $\varepsilon_q = 0$ that step is many orders of magnitude below any change the orbit can register, so a direct 14-parameter fit cannot move ε_q away from its starting value. The thesis' quoted best fit, $\varepsilon_q = 0$, is that starting value. Its interval comes from the profile and is unaffected.

The extended mass. Stars, stellar remnants or dark matter around Sgr A* would add a spherical mass distribution. We model it, as GRAVITY Collaboration

(2020) did, as a Plummer (1911) sphere with scale radius $0.3''$. It enters the metric through its Newtonian potential in g_{tt}; its own post-Newtonian terms are smaller by a further $O(M/r)$. Two unit tests guard the implementation. Switched off, the model is bit-identical to the thesis model. And a sphere ten times wider than the orbit reproduces, to within 3 per cent, the textbook retrograde precession of a homogeneous core, $\Delta\omega = -3\pi f (a/b)^3 \sqrt{1-e^2}$ per orbit. As in GRAVITY Collaboration (2022), its size is quoted as the fraction of M enclosed within S2's apocentre ($0.235''$), which holds 23.5 per cent of the sphere's mass.

7.2 The no-hair reach grows faster than $1/\sqrt{N}$

The thesis extrapolated its no-hair bound by scaling the uncertainty as $1/\sqrt{N}$ with the number of orbits. Ten further orbits of existing quality would then shrink $\sigma(\varepsilon_q)$ by a factor 2.8, to 34 000. The Fisher forecast for exactly that data set gives $\sigma(\varepsilon_q) = 577$, a gain of 162 (Fig. 7.1, left).

The difference is structural. ε_q acts on S2 almost only through a *secular* precession of its pericentre, and the Gillessen et al. (2017) data contain a single pericentre passage (2002). Within one orbit, the orientation elements absorb most of the quadrupole's effect. Each further passage measures the accumulated advance directly, and a drift measured over a baseline T improves as $T^{-3/2}$, not $T^{-1/2}$. The jumps in the existing-quality curve sit at the pericentre passages. The $1/\sqrt{N}$ figure in the thesis is therefore a strong underestimate of what monitoring can achieve.

GRAVITY-era astrometry (65 μas per coordinate at 18 epochs per year, against 0.4 mas at 9.2 for the adaptive-optics data) gains faster still: $\sigma(\varepsilon_q)$ falls to 5 600 after 8 years, 3 700 after 16 (one orbit), 840 after 32 and 290 after 64. That is two orders of magnitude within a few decades. It is still far from the Kerr quadrupole itself, $|\tilde{q}| = a_*^2 < 1$ in units of M^3, so S2 is too far out to test the no-hair theorem at that level, as the thesis argued.

7.3 Frame dragging

At $a_* = 0.94$, frame dragging shifts S2's pericentre by only -0.204 arcmin per orbit, against 11.98 arcmin for the Schwarzschild advance (both from the thesis' S2 fit). GRAVITY measured the Schwarzschild advance with a 19 per cent uncertainty (GRAVITY Collaboration, 2020). With a_* free in the Kerr model

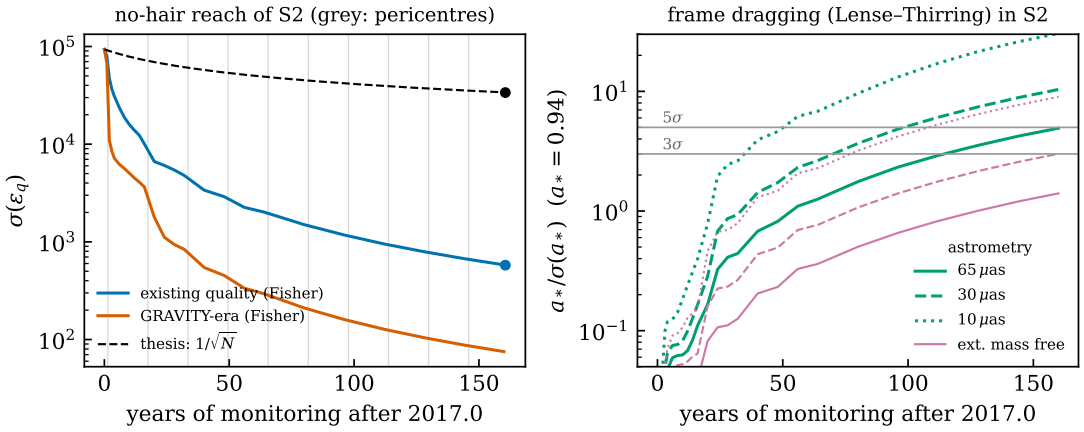


Figure 7.1: Left: forecast uncertainty of the no-hair parameter versus monitoring time after 2017.0, for the two continuations of the data (grey lines: S2's pericentre passages). The dashed curve is the thesis' $1/\sqrt{N}$ extrapolation; the dots compare it with the Fisher forecast at ten more orbits. Right: significance of the spin of Sgr A* ($a_* = 0.94$, Lense–Thirring and Kerr quadrupole) for three astrometric precisions per coordinate (green; purple: with an extended mass fitted at the same time). Spin axis along the line of sight.

(Fig. 7.1, right), the spin reaches 3σ after 114 yr of monitoring at 65 μas, and 5σ after > 160 yr. Better astrometry helps strongly: at 30 μas the times are 70 and 98 yr, and at 10 μas they are 34 and 50 yr.

These times assume that nothing else precesses the orbit. If an extended mass is fitted at the same time, they become > 160 and > 160 yr at 65 μas, 160 and > 160 yr at 30 μas, and 77 and 108 yr at 10 μas. This is the confusion that Merritt et al. (2010) identified, and that GRAVITY Collaboration (2020) noted for their own extended-mass limit. The direction of the spin axis matters: here it lies along the line of sight, as throughout the thesis.

7.4 The extended mass and the no-hair parameter

An extended mass and a positive ε_q both advance S2's pericentre retrogradely, so the two parameters trade against each other. With both free, their correlation is -0.79 on the Gillessen et al. (2017) data, and -0.45, -0.60 and -0.58 after 8, 16 and 32 yr of GRAVITY-era monitoring. The extended mass itself is constrained

to $\sigma = 0.854$ per cent of M within the apocentre by the Gillessen et al. (2017) data alone, and to 0.076 per cent after 8 yr of GRAVITY-era data. The second figure is consistent with the ≈ 0.1 per cent that GRAVITY achieved with three years of such data (GRAVITY Collaboration, 2020, 2022). Fitting the extended mass jointly costs little: it widens $\sigma(\varepsilon_q)$ by factors of only 1.12, 1.25 and 1.23 at 8, 16 and 32 yr (Fig. 7.2, left).

Leaving it out is a different matter (Table 7.1, Fig. 7.2, right). Take an extended mass at GRAVITY's present 1σ bound, 0.1 per cent of M within the apocentre, fitted with the pure Kerr model. It produces a false no-hair signal of 0.5σ after 8 yr, 1.4σ after 16 yr and 5.0σ after 32 yr; half that mass still gives 2.5σ at 32 yr. Here σ is the uncertainty the biased fit itself would quote, from the curvature of its profile. The mass is biased only slightly, by $+0.32\sigma$, $+0.42\sigma$ and $+0.35\sigma$.

Nothing in the fit gives the problem away. Measured against the true Kerr orbit, the extended mass alters the data by $\chi^2 = 59\,000$, 86 000 and 390 000, far more than their errors, but the orbit elements absorb nearly all of it. The χ^2 the Kerr fit leaves over is only 1.6, 5.3 and 73. Noise alone scatters the χ^2 of these $N = 662$, 990 and 1646 data values by $\sqrt{2N} = 36$, 44 and 57, so the leftover is lost in it.

That absorption also defeats linearised estimates. It moves the orbit a long way along its best-measured direction, where the model is no longer linear. The linear response about the true orbit recovers only 0.52 of the fitted 32-yr ε_q bias, and for some data sets it gets the mass bias wrong in sign. Every bias quoted here therefore comes from a full refit.

So any future no-hair test with S2 must fit an extended mass jointly. Doing so is cheap. Not doing so turns the extended mass, which can be no larger than the present bound, into the leading systematic error. The thesis named this risk in its discussion; these numbers quantify it.

Table 7.1: An extended mass (Plummer, scale 0.3″) present in the data but left out of the fit: biases of the 13-parameter Kerr fit in M and R_0 and the χ^2 it leaves, and the false no-hair signal in the thesis' ε_q profile. Noise-free data on the Gillessen et al. (2017) epochs, plus the stated years of GRAVITY-era monitoring. For M and R_0, σ is the forecast uncertainty of the same data set; for ε_q it is the uncertainty the biased fit itself would quote, from the curvature of its profile.

M_{ext} [% of M]	data	$\Delta M/\sigma$	$\Delta R_0/\sigma$	$\Delta\epsilon_q$	$\Delta\epsilon_q/\sigma$	χ^2_{left}
0.05	Gillessen 2017	+0.02	+0.02	2 550	+0.03	0.0034
0.05	+ 8 yr	+0.16	+0.18	1 450	+0.24	0.41
0.05	+ 16 yr	+0.20	+0.22	2 550	+0.68	1.3
0.05	+ 32 yr	+0.14	+0.25	2 090	+2.48	18
0.10	Gillessen 2017	+0.05	+0.03	6 830	+0.07	0.014
0.10	+ 8 yr	+0.32	+0.37	2 880	+0.47	1.6
0.10	+ 16 yr	+0.42	+0.45	5 120	+1.36	5.3
0.10	+ 32 yr	+0.35	+0.56	4 180	+4.98	73
0.20	Gillessen 2017	+0.09	+0.06	1.5×10^4	+0.17	0.055
0.20	+ 8 yr	+0.64	+0.75	5 780	+0.94	6.4
0.20	+ 16 yr	+0.89	+0.92	1.0×10^4	+2.78	22
0.20	+ 32 yr	+0.71	+1.13	8 360	+9.99	300

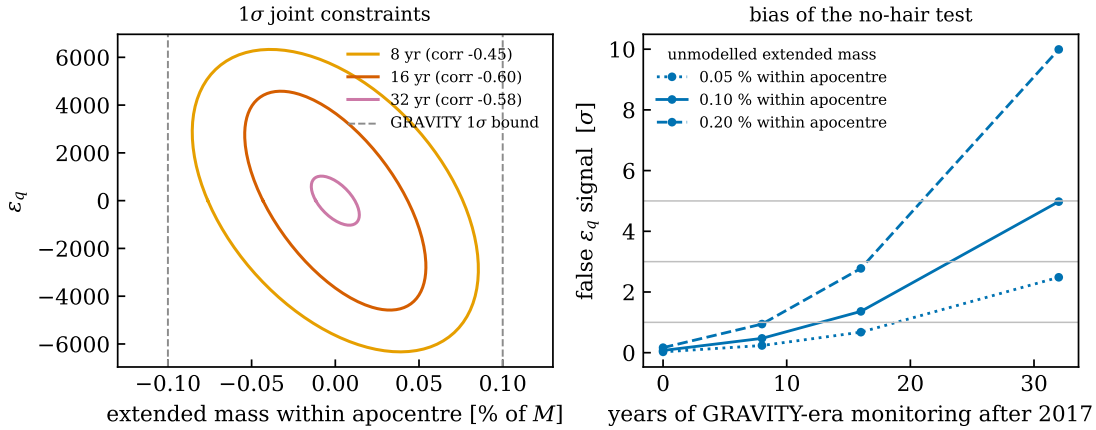


Figure 7.2: Left: joint 1σ constraints on the extended mass enclosed within S2's apocentre and the no-hair parameter, after 8, 16 and 32 yr of GRAVITY-era monitoring (grey: GRAVITY's present 1σ bound). Right: the false no-hair signal, in units of the forecast $\sigma(\varepsilon_q)$, when the extended mass is left out of the fit.

Chapter 8

Falling into Sagittarius A*

The last capability of the simulator is the one most often asked of a black hole: what would someone falling in see and feel? None of it is a matter of taste. The light, the clocks and the tides follow from the metric, and this chapter computes each of them for Sgr A* ($a_* = 0.94$) along one physical worldline. The fall is rendered live with every effect on.

8.1 The trip

Worldline. A free fall is a timelike geodesic of the same Hamiltonian the ray tracer uses, $H = \frac{1}{2} g^{\mu\nu} p_\mu p_\nu$, with $H = -\frac{1}{2}$ in place of 0, so the affine parameter is the proper time τ of the faller. It is integrated in double precision (Dormand–Prince, tolerance 10^{-12}) in the *ingoing* Kerr–Schild chart (Chapter 3), which is regular across the future horizon where Boyer–Lindquist coordinates fail. The default start is *rain*: an observer dropped from rest at infinity ($E = 1$, $L_z = 0$, $Q = 0$), whose polar angle stays constant, the "river" frame of Hamilton & Lisle (2008). The trip begins at $r_0 = 30\,M$ at a polar angle of 60°. A release from rest relative to the local ZAMO is the alternative start.

Where it stops. The trip ends just above the inner (Cauchy) horizon, at $r = 0.709\,M$ ($r_- = 0.659\,M$). Perturbations falling in pile up at the inner horizon and drive the mass inflation of Poisson & Israel (1990), so the classical Kerr interior beyond it is not a reliable description of a real black hole. The simulator says so on screen rather than render it.

Checks. In Schwarzschild the worldline reproduces the closed forms for both starts: the cycloid of a fall from rest, and $\tau = \frac{2}{3}(r_0^{3/2} - r^{3/2})/\sqrt{2M}$ for rain. The proper time to $r = 2M$ matches them to 1×10^{-14}, and unit tests check radii

inside the horizon as well. For $a_* = 0.94$, across the 350 steps of the default trip through r_+, L_z and Q are conserved to 2×10^{-13} and 3×10^{-13}, and the norm of the 4-velocity to 1×10^{-12}; $E = -p_t$ is constant by construction. Q is reconstructed through Boyer–Lindquist components, so it is checked away from the two horizons, and it uses Carter's constant for timelike geodesics. The tracer's existing routine assumed light; this chapter generalised it.

Camera. The faller's frame takes $e_0 = u$ and builds its three spatial legs by Gram–Schmidt, with the full metric, from the Kerr–Schild coordinate directions ∂_r, ∂_θ, ∂_ϕ projected orthogonal to u. Those directions are smooth through r_+ in this chart, so the camera is too, and the legs mirror the exterior observers' triad, so looking around works exactly as in Chapter 4.

Light. The live renderer traces the same backward rays as before, switched to the ingoing chart. One thing changes: what "captured" means. Traced backward from inside r_+, every ray leaves outward through the future horizon, because the light came in from outside. A backward ray that heads inward from outside, by contrast, is bound for the *past* horizon, which this chart reaches only as $t \to -\infty$; such rays carry no light in this model. The CPU twin applies the same rule. On 8 image strips through the centre of the faller's view, at radii from $6\,M$ to $0.9\,M$ (two of them inside the horizon), the edges of the dark region in the live image agree with the double-precision tracer within 0.73 pixels. Across the horizon itself the image evolves smoothly. Between full frames at $r_+ + 0.03\,M$ and $r_+ + 0.01\,M$, 0.35 per cent of the pixels change between dark and sky; across r_+, from $r_+ + 0.01\,M$ to $r_+ - 0.01\,M$, 0.33 per cent; and inside, from $r_+ - 0.01\,M$ to $r_+ - 0.03\,M$, 0.32 per cent.

8.2 Time

From r_0, the faller reaches the ergosphere ($r_E = 1.883\,M$ at this latitude) after $\tau = 76.08\,M$ of proper time, and the horizon after $76.54\,M$: 27.0 minutes for Sgr A*, whose unit of time is $GM/c^3 = 21.2\,\mathrm{s}$. Only 9.6 s pass between entering the ergosphere and crossing the horizon. From the horizon to the end of the model just above r_- takes $0.403\,M$, or 8.5 s.

Far-away time behaves very differently (Fig. 8.2, left). The Boyer–Lindquist time t of a distant clock grows without bound as the faller approaches r_+, so a distant observer never sees the crossing. The faller's own clock notices nothing, and the Kerr–Schild time stays finite. Inside, t is no longer a time at all: no signal

sent from there can reach the outside.

8.3 What you see

Figure 8.1 shows the view straight ahead, at the hole, with the Milky Way panorama as the distant sky (the hypothetical thin disk is switched off: Sgr A* has none). Far out, the shadow sits in the lensed Milky Way. As the faller speeds up, aberration and the Doppler shift crowd the sky toward the direction of motion and blueshift it, and the shadow swells until, near r_+, it fills most of the view ahead. Crossing the horizon is a non-event locally. The frames on either side of it differ smoothly, with no seam or flash, as the regular chart demands. Inside, the outside universe is still there. Behind the faller it fills the whole view, redshifted and dimmed (below). Ahead it survives only as a band at the edge of the view. The dark region ahead is light that would come from the past horizon, from no source in this model.

The colour of that light changes along the way, and it depends on direction. Ahead, the light that skims the edge of the dark region is blueshifted: it arrives at 2.0–2.2 times its frequency at infinity at $6\,M$, up to 5.6 times at $2\,M$, and up to 27.3 times at $0.9\,M$, inside the horizon. Behind, the faller is running away from the light, and it is redshifted: it arrives at 0.79 times its frequency at infinity at the start, 0.47 times at the horizon and 0.42 times at the end of the trip. Light from the side arrives at 0.85 times its frequency at infinity at the horizon (Fig. 8.2, middle).

8.4 Tides

The tidal field felt by the faller is the electric part of the Riemann tensor in its own frame, $E_{ij} = R_{i\alpha j\beta}u^\alpha u^\beta$: two free particles separated by ξ accelerate apart at $-E_{ij}\xi^j$. The simulator computes it exactly from the analytic metric. The Christoffel symbols come from complex-step derivatives, and their derivatives from central differences. It is checked in two ways. In Schwarzschild it reproduces the radial-fall values $-2M/r^3$ and M/r^3, also inside the horizon. And everywhere on the trip $8(E_{ij}E^{ij} - B_{ij}B^{ij})$ equals the closed-form Kretschmann scalar of Kerr to 8×10^{-9}.

Sgr A* is gentle. Across a 2 m body, the tidal stretch is $3.7 \times 10^{-3}\,\mathrm{m\,s^{-2}}$ at the horizon and $2.8 \times 10^{-2}\,\mathrm{m\,s^{-2}}$ at the end of the trip (Fig. 8.2, right). Tides scale as

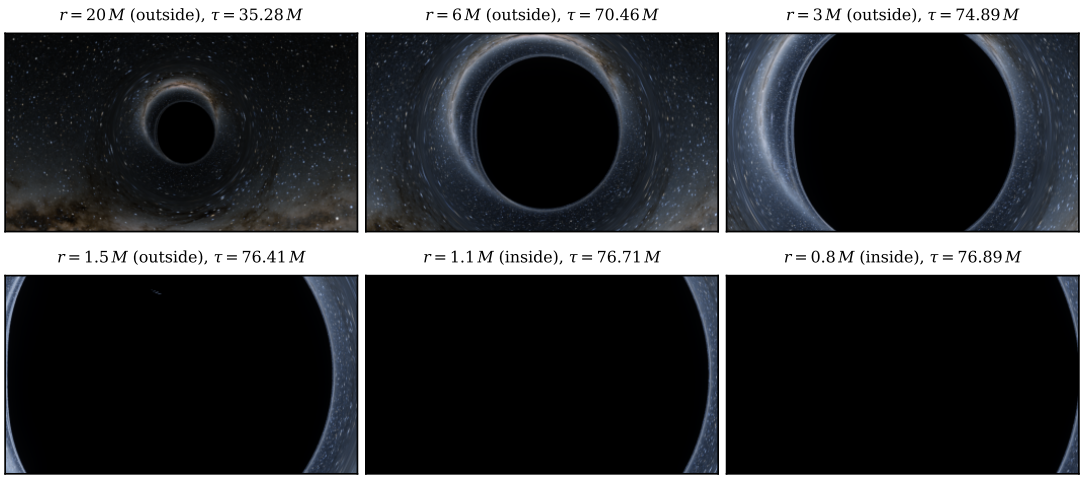


Figure 8.1: Falling into Sgr A* ($a_* = 0.94$): the view straight ahead of an observer dropped from rest at infinity, at polar angle 60°, live GPU renders of the double-precision worldline (field of view 1.6 rad; the lensed Milky Way is the only light). Top: outside the horizon ($r_+ = 1.341\,M$); bottom: just outside it and inside it. Labels give the radius and the faller's proper time since $r_0 = 30\,M$.

M/r^3, and the horizon radius scales with M, so at the horizon they fall as $1/M^2$. A black hole of four million solar masses can be entered intact; a stellar-mass one could not be.

8.5 Limits

The fall is classical, vacuum Kerr, along one geodesic. The accretion flow, the hypothetical disk and every quantum effect are left out, and the trip stops at the inner horizon by design. The sky is the panorama seen from Earth, used as the distant background, as in the rest of this report. The camera frame is one choice among many: another tetrad rotates or boosts the view but changes no physics.

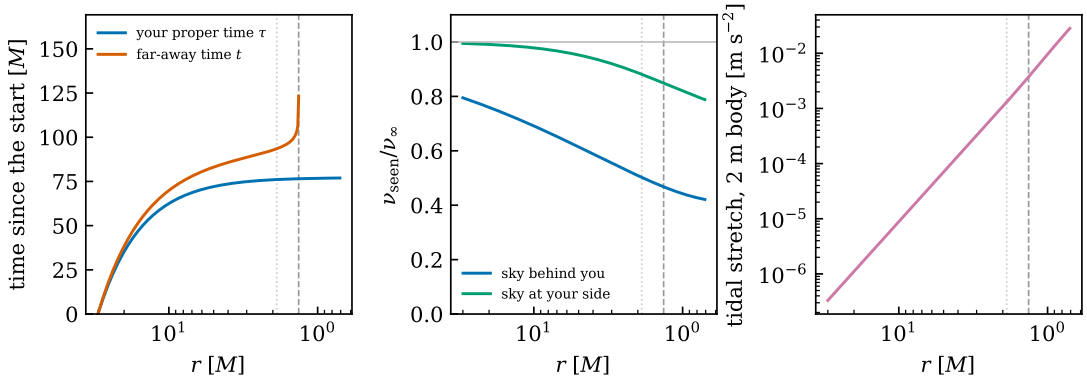


Figure 8.2: The faller's instruments along the trip (dashed: the horizon; dotted: the ergosphere at this latitude). Left: its proper time and the far-away time, which diverges at r_+. Middle: frequency of the sky light arriving from straight behind and from the side, relative to its frequency at infinity. Right: tidal stretch across a 2 m body.

Chapter 9

Discussion and Limitations

The chapters so far share a pattern: each new capability is first a double-precision tool tested against closed forms, then a live rendering path compared with that tool. The comparisons are what make the images defensible, and they have found real defects. Three were in the renderer: the double-counted colour, the step budget and the missing constraint projection. Two were in the orbit fitting. The forecast Jacobian needs finite-difference steps far smaller than the thesis used, and the thesis' direct no-hair fit cannot move ε_q off its starting value (Chapter 7). Several limitations remain and are stated where they arise.

- *The thin disk is hypothetical.* At Sgr A*'s measured accretion rate a thin, radiatively efficient disk would be 12–1 200 times too luminous (§2.5). The renders that show one are labelled as such; the physical flow is the subject of Chapter 6.

- *The sky is Earth's.* The Milky Way panorama is the sky seen from the Solar System, used as a backdrop at infinity, and its colour shift treats each pixel as a 6500 K blackbody (§2.6).

- *The hot spot is the simplest model.* A rigid Gaussian on a circular geodesic, optically thin, with a fixed spectral index. It reproduces the GRAVITY periods and sense of rotation but not the size of the observed centroid motion, exactly the tension GRAVITY reported. Shear, cooling, out-of-plane motion and polarisation are not modelled (Chapter 5).

- *Display processing is not physics.* Bloom, tone curves, white balance and false colour are confined to the display chain and are switched off in every quantitative comparison (§2.4).

- *The accretion flow is a semi-analytic RIAF, not a simulation.* Its radial profiles are fixed power laws, its electrons are purely thermal, and its one free parameter is fitted to a single flux density. The comparison with the EHT ring is therefore a test of this family of models, not of general relativity. In that test the family's reference temperature fails: the ring is too small. The thesis' critical-curve metrology, which needs no emission model, remains the result that tests the spacetime.

- *Polarisation without Faraday effects.* Rotation and conversion in the flow are omitted, and the intrinsic polarisation fraction is a fixed assumption. Only quantities independent of that fraction — the phase of β_2 and $|\beta_2|/|\beta_1|$ — are compared with the EHT. Interstellar scattering toward Sgr A* is not modelled; the EHT notes that it barely affects β_2 (Event Horizon Telescope Collaboration et al., 2024a).

- *The forecasts are idealised.* They use noise-free data, sampled uniformly with no seasonal gaps, and include no reference-frame systematics, source confusion or second star. They are local (Fisher) estimates, checked against full profile-likelihood fits to within 8 per cent. The extended mass is a single Plummer profile at GRAVITY's scale radius, and the spin axis lies along the line of sight. Both choices change the numbers, though not the conclusion that an extended mass must be fitted with any future no-hair or spin test (Chapter 7).

- *The wormhole is not Sgr A*.* The Dneg metric is a test of the renderer's generality. The exotic matter it would need is not modelled, its other universe's sky is an illustrative recolouring, and nothing about the existence of such objects is implied (Appendix A).

- *The fall is classical and stops at r_-.* It is one geodesic in vacuum Kerr: no accretion flow, no quantum effects, and no interior beyond the inner horizon, where mass inflation makes the classical solution unreliable (Chapter 8).

Chapter 10

Conclusions

The simulator now shows Sgr A* as an observer would see it and keeps the thesis' standard of evidence. The main results so far:

- A thin disk fed at the measured accretion rate would be a blue-white disk at 7 280–23 000 K, and 12–1 200 times brighter than Sgr A* is: the accretion flow must be radiatively inefficient.

- A moving observer's view follows from a Lorentz-boosted tetrad alone, and is verified against Synge's shadow plus aberration to 2×10^{-12} and in the live renderer to 0.72 pixel.

- On the orbits GRAVITY fitted to the 2018 flares, a hot spot traced with slow light reproduces the period and clockwise sense of the observed loops, moves only half as far as observed in 30 minutes, and echoes every flash 5.9–9.1 minutes later in its lensed image.

- A radiatively inefficient flow normalised to the EHT's 2.4 Jy puts its 230 GHz light in a crescent inside the photon ring. Its ring is 17 per cent smaller than the EHT's unless its electrons are cooler. Its polarisation reproduces the EHT's spiral only with a vertical, poloidal magnetic field.

- Continued monitoring of S2 tightens the no-hair parameter much faster than the thesis' $1/\sqrt{N}$ extrapolation: ten more orbits of existing quality reach $\sigma(\varepsilon_q) = 577$, not 34 000. An extended mass at GRAVITY's present bound, if left out of the fit, turns into a false no-hair signal of 5.0σ after 32 years. Fitted jointly, it costs a factor of only 1.23 in $\sigma(\varepsilon_q)$.

- Frame dragging in S2's orbit reaches 3σ after 34 yr at $10\,\mu$as astrometry, and after 114 yr at today's $65\,\mu$as. With an extended mass fitted alongside, these become 77 yr and > 160 yr.
- The renderer is not tied to Kerr. On the Interstellar wormhole, its live image reproduces the exact edge of the other universe within 0.28 pixel.
- Falling into Sgr A* from $30\,M$ takes 27.0 minutes of the faller's time to the horizon and 8.5 s more to the inner horizon. It is gentle: the tidal stretch across a 2 m body is $3.7 \times 10^{-3}\,\mathrm{m\,s^{-2}}$ at the horizon. The live view is checked against the double-precision tracer, inside the horizon as well as outside, to within 0.73 pixels.

Appendix A

An Exotic Spacetime: the Traversable Wormhole

Nothing in this appendix describes Sgr A.* It applies the simulator to a hypothetical spacetime, the Double Negative ("Dneg") wormhole designed for the film *Interstellar* by James et al. (2015) after the traversable wormholes of Morris & Thorne (1988). The point is to test the code's generality: the same camera, sky and validation machinery, applied to a metric that is not Kerr. In the simulator it is reached only through a separate menu entry, EXOTIC (NOT SGR A*). A banner on screen says the same, and none of its numbers enter any Sgr A* result.

A.1 The metric and its rays

The Dneg wormhole without gravity (James et al., 2015, eqs. 1 and 5) is

$$ds^2 = -dt^2 + d\ell^2 + r(\ell)^2\,(d\theta^2 + \sin^2\theta\, d\phi^2), \tag{A.1}$$

$$r(\ell) = \begin{cases} \rho & |\ell| \le a, \\ \rho + M\left[x\arctan x - \tfrac{1}{2}\ln(1+x^2)\right], \quad x = \dfrac{2(|\ell|-a)}{\pi M} & |\ell| > a. \end{cases} \tag{A.2}$$

The proper radial distance ℓ runs from $-\infty$ (the other universe) through the cylindrical throat of radius ρ and length $2a$ to $+\infty$ (our universe). The parameter M sets how gently the mouths flare out. James et al. (2015) describe the flare by the *lensing width* $W = 1.42953\,M$. It is the lateral distance in the embedding diagram over which the surface turns from vertical to 45°, which is where $dr/d\ell = 1/\sqrt{2}$. Our code finds that point by bisection and reproduces their coefficient: the numerical W and the closed form agree to 9×10^{-16}. We use the film's wormhole

throughout, as chosen by Christopher Nolan (James et al., 2015): $2a = 0.01\,\rho$ and $W = 0.05\,\rho$ ($M = 0.03498\,\rho$). The weak Newtonian potential that James et al. (2015) added for the spacecraft's trajectory, $|\Phi| \lesssim 10^{-12}$, is neglected for light, as in their own ray tracing.

With $p_t = -1$ the affine parameter is the time t. Two quantities are conserved: $b = p_\phi$, and the total impact parameter $B^2 = p_\theta^2 + b^2/\sin^2\theta$. The rays obey their equations (A.7):

$$\frac{d\ell}{dt} = p_\ell, \quad \frac{d\theta}{dt} = \frac{p_\theta}{r^2}, \quad \frac{d\phi}{dt} = \frac{b}{r^2\sin^2\theta}, \quad \frac{dp_\ell}{dt} = B^2\,\frac{dr/d\ell}{r^3}, \quad \frac{dp_\theta}{dt} = \frac{b^2\cos\theta}{r^2\sin^3\theta}. \tag{A.3}$$

The fp64 reference (`science/wormhole`) integrates eq. (A.3) backward from the camera with an adaptive Dormand–Prince scheme (tolerance 10^{-12}). The camera uses their local basis $e_x = e_{\hat\ell}$, $e_y = e_{\hat\phi}$, $e_z = -e_{\hat\theta}$. Along every traced ray $|\Delta B^2|/B^2 \le 4\times10^{-16}$ and the null condition $|p_\ell^2 + B^2/r^2 - 1| \le 7\times10^{-12}$.

The live renderer uses the fact that, by spherical symmetry, each ray moves in a plane through the centre (its lateral motion follows a great circle; James et al., 2015). It therefore integrates the planar form $d\ell/dt = p_\ell$, $d\psi/dt = B/r^2$, $dp_\ell/dt = B^2 r'/r^3$ in fp32 RK4. Here ψ is the angle swept from the camera's position, and the polar singularity of eq. (A.3) never arises. Both integrations stop once the ray is far out and receding ($|\ell| \ge 400\,\rho$ on the CPU, $40\,\rho$ on the GPU). The remaining sweep, $\arcsin(B/r)$, is then added as a straight line. Out there $dr/d\ell = 1 - O(M/\ell)$, so that completion errs by $O(BM/\ell^2)$: moving the CPU stopping point from $400\,\rho$ to $4000\,\rho$ shifts the Einstein rings below by at most 3×10^{-6} degrees.

A.2 The edge of the other universe

The radial equation has an exact first integral, $p_\ell^2 = 1 - B^2/r(\ell)^2$. Since r is smallest, $r = \rho$, along the whole throat, a ray gets through if and only if $B < \rho$. A ray leaving the camera at angle α from the direction of the wormhole has $B = r(\ell_c)\sin\alpha$. The other universe therefore fills a disk on the camera's sky whose edge is exactly at

$$\sin\alpha_{\text{edge}} = \rho/r(\ell_c), \tag{A.4}$$

whatever a and M are. This is our own elementary result; we use it as an exact test of both tracers. Bisecting the CPU tracer for the edge reproduces eq. (A.4) to 7×10^{-12} degrees. The live GPU path is checked on 2001-pixel strips along

Table A.1: The Dneg wormhole of *Interstellar* ($2a = 0.01\,\rho$, $W = 0.05\,\rho$) seen from three distances: angular radius of the other universe (CPU, equal to eq. A.4) and of the Einstein ring. The live GPU edges agree with the CPU edges within 0.28 pixels in all three cases.

ℓ_c/ρ	$r(\ell_c)/\rho$	edge [deg]	Einstein ring [deg]
15	15.764	3.637	5.428
6	6.796	8.462	10.120
2	2.834	20.659	21.662

both image axes through the centre, using a validation mask (rays that end in our universe are white, rays that end in the other universe are black). All four edges (left, right, top, bottom) lie within 0.28 pixels of the CPU value at every camera distance (Table A.1).

Rays with B just above ρ wind around the throat before returning to our universe, and they image our own sky many times in thin bands around the edge. In every direction around the wormhole, exactly one of these rays ends at the point of our celestial sphere directly behind the wormhole. Together they form the *Einstein ring*, the ring image of a source exactly behind the wormhole (James et al., 2015). Its radius on the camera's sky is found by bisecting for the ray whose final azimuth is $\phi_\infty = \pi$ (the camera sits at $\phi = 0$).

A.3 The live demo

The demo shows our universe as the Milky Way map of Chapter 2. For the other universe, whose sky nobody can know, it shows the same map turned around with its colour channels swapped, so the dust lanes turn blue and it cannot be mistaken for ours. This choice is illustrative only and is labelled as such on screen. Dragging the mouse moves the camera around the wormhole, as it does around Sgr A*, and the scroll wheel flies along ℓ, through the throat if you keep going. No colour shift is applied, because none exists: $g_{tt} = -1$ everywhere, so a static camera receives light from either universe at the frequency it was sent. Figure A.1 shows the approach. The sine of the other universe's angular radius grows as $1/r(\ell_c)$. Around its edge, inside the Einstein ring, lies the strongly lensed image of our own Milky Way, with the Galactic Centre directly behind the throat. This is the structure James et al. (2015) describe for the film's wormhole. The far universe

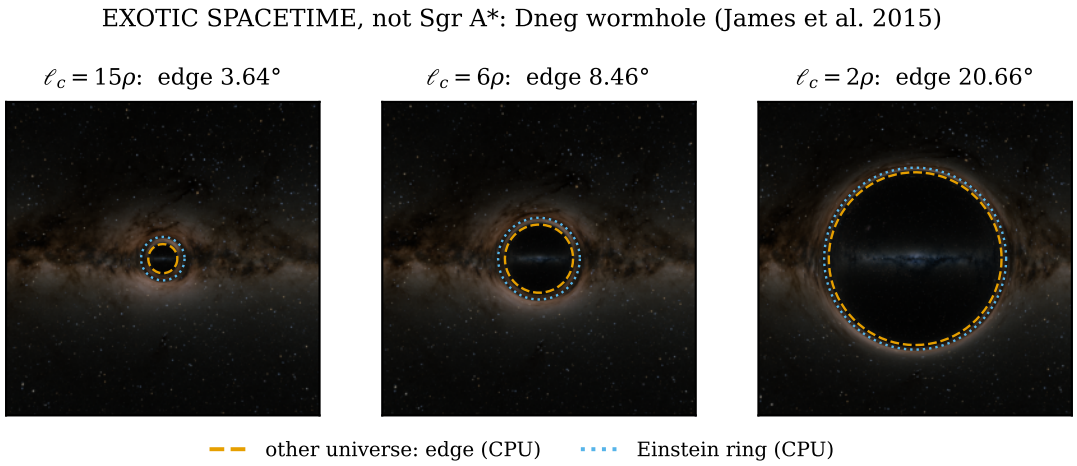


Figure A.1: *Not Sgr A*:* the Interstellar wormhole rendered live (GPU), approached along ℓ (field of view 1.2 rad). The fp64 CPU tracer's results are overlaid: the edge of the other universe (dashed; eq. A.4) and the Einstein ring (dotted). Inside the edge is the other universe (an illustrative recoloured sky); outside it is the lensed Milky Way, with the Galactic Centre behind the throat.

is seen through the mouth. The star field on the camera's side forms secondary lensed images near the mouth's edge, which are very thin for a wormhole this short. The Einstein ring lies just outside the edge.

Appendix B

Reproducing This Report

`make reproduce-ext` runs, in order, every tool whose output this report quotes, then regenerates the numbers and figures:

Command	Produces
`ctest --preset sci-rel`	all unit tests (thesis and extension)
`spectral_tool`	`spectral.json`: thin disk at the measured $\dot{M}$; visual vs bolometric contrast
`compare_live_colour.py`	`g13_colour.json`: live colour vs CPU reference
`observer_tool`	`observers.json`: observer kinematics and shadow edges
`compare_observers.py`	`g17_observers_gpu.json`: live observers vs fp64
`flare_tool`	`flares.json`, light curves, centroid tracks, echoes, images
`compare_flare.py`	`g18_flare_gpu.json`: live hot spot vs fp64
`riaf_tool`	`riaf.json`: 230 GHz flow, fit, ring, visibilities, sensitivity, polarisation
`compare_riaf.py`	`g19_riaf_gpu.json`: live 230 GHz band vs fp64
`forecast_tool`	`forecasts.json`: Fisher forecasts and extended-mass biases (reads the thesis fit, `validation/phase5`)
`wormhole_tool`	`wormhole.json`: the exotic Dneg wormhole (not Sgr A*)
`compare_wormhole.py`	`wormhole_gpu.json`: live wormhole edges vs fp64
`infall_tool`	`infall.json`: the fall into Sgr A* (worldline, light, tides, reference strips)
`compare_infall.py`	`g24_infall_gpu.json`: live in-fall view vs fp64, continuity through r_+
`render_ext_gallery.py`	engine renders used as figures
`make_flare_movie.py`	one-orbit flare movie (GIF)
`make_numbers_ext.py`	`numbers_ext.tex`, `tab_*.tex`
`make_figures_ext.py`	`figures/`

Tools write to `validation/extension/`; the GPU comparisons need the Vulkan build (`cmake --build --preset engine-rel`) and run with the validation layers on, which must report zero errors.

Bibliography

Bardeen J. M., Press W. H., Teukolsky S. A., 1972, The Astrophysical Journal, 178, 347

Broderick A. E., Loeb A., 2006, Monthly Notices of the Royal Astronomical Society, 367, 905

Broderick A. E., Fish V. L., Doeleman S. S., Loeb A., 2011, The Astrophysical Journal, 735, 110

Carter B., 1968, Physical Review, 174, 1559

Chael A., Johnson M. D., Lupsasca A., 2021, The Astrophysical Journal, 918, 6

Connors P. A., Piran T., Stark R. F., 1980, The Astrophysical Journal, 235, 224

Dolence J. C., Gammie C. F., Mościbrodzka M., Leung P. K., 2009, The Astrophysical Journal Supplement Series, 184, 387

Event Horizon Telescope Collaboration 2022, The Astrophysical Journal Letters, 930, L12

Event Horizon Telescope Collaboration Akiyama K., Alberdi A., et al., 2024a, The Astrophysical Journal Letters, 964, L25

Event Horizon Telescope Collaboration Akiyama K., Alberdi A., et al., 2024b, The Astrophysical Journal Letters, 964, L26

GRAVITY Collaboration 2018, Astronomy and Astrophysics, 615, L15

GRAVITY Collaboration 2020, Astronomy and Astrophysics, 636, L5

GRAVITY Collaboration 2022, Astronomy and Astrophysics, 657, L12

GRAVITY Collaboration Abuter R., Amorim A., Bauböck M., Berger J. P., et al., 2018, Astronomy and Astrophysics, 618, L10

GRAVITY Collaboration Bauböck M., Dexter J., Abuter R., Amorim A., et al., 2020, Astronomy and Astrophysics, 635, A143

Gaia Collaboration Brown A. G. A., Vallenari A., et al., 2018, Astronomy and Astrophysics, 616, A1

Genzel R., Eisenhauer F., Gillessen S., 2010, Reviews of Modern Physics, 82, 3121

Gillessen S., et al., 2017, The Astrophysical Journal, 837, 30

Hamaus N., Paumard T., Müller T., Gillessen S., Eisenhauer F., Trippe S., Genzel R., 2009, The Astrophysical Journal, 692, 902

Hamilton A. J. S., Lisle J. P., 2008, American Journal of Physics, 76, 519

Himwich E., Johnson M. D., Lupsasca A., Strominger A., 2020, Physical Review D, 101, 084020

James O., von Tunzelmann E., Franklin P., Thorne K. S., 2015, American Journal of Physics, 83, 486

Jimenez J., 2014, Next Generation Post Processing in Call of Duty: Advanced Warfare, Advances in Real-Time Rendering in Games, SIGGRAPH 2014 course, `https://advances.realtimerendering.com/s2014/`

Kerr R. P., 1963, Physical Review Letters, 11, 237

Kerr R. P., Schild A., 2009, General Relativity and Gravitation, 41, 2485

Leung P. K., Gammie C. F., Noble S. C., 2011, The Astrophysical Journal, 737, 21

Marrone D. P., Moran J. M., Zhao J.-H., Rao R., 2007, The Astrophysical Journal, 654, L57

Merritt D., Alexander T., Mikkola S., Will C. M., 2010, Physical Review D, 81, 062002

Misner C. W., Thorne K. S., Wheeler J. A., 1973, Gravitation. W. H. Freeman

Morris M. S., Thorne K. S., 1988, American Journal of Physics, 56, 395

Narkowicz K., 2016, ACES Filmic Tone Mapping Curve, `https://knarkowicz.wordpress.com/2016/01/06/aces-filmic-tone-mapping-curve/`

Page D. N., Thorne K. S., 1974, The Astrophysical Journal, 191, 499

Palumbo D. C. M., Wong G. N., Prather B. S., 2020, The Astrophysical Journal, 894, 156

Plummer H. C., 1911, Monthly Notices of the Royal Astronomical Society, 71, 460

Poisson E., Israel W., 1990, Physical Review D, 41, 1796

Roberts M., 2018, The Unreasonable Effectiveness of Quasirandom Sequences, `https://extremelearning.com.au/unreasonable-effectiveness-of-quasirandom-sequences/`

Rybicki G. B., Lightman A. P., 1979, Radiative Processes in Astrophysics. Wiley, New York

Sobotka T., 2022, AgX: a display rendering transform, `https://github.com/sobotka/AgX`

Synge J. L., 1966, Monthly Notices of the Royal Astronomical Society, 131, 463

Visser M., 2007, The Kerr spacetime: A brief introduction, arXiv:0706.0622; introductory chapter of *The Kerr Spacetime*, eds. S. Scott, M. Visser and D. Wiltshire (Cambridge University Press)

Walker M., Penrose R., 1970, Communications in Mathematical Physics, 18, 265

Wrensch B., 2023, Minimal AgX Implementation, Iolite engine development blog, `https://iolite-engine.com/blog_posts/minimal_agx_implementation`

Wright E., 2020, Deep Star Maps 2020, NASA/Goddard Space Flight Center Scientific Visualization Studio, `https://svs.gsfc.nasa.gov/4851`; Gaia DR2: ESA/Gaia/DPAC

Wyman C., Sloan P.-P., Shirley P., 2013, Journal of Computer Graphics Techniques, 2, 1

Yuan F., Quataert E., Narayan R., 2003, The Astrophysical Journal, 598, 301